Passive Fault Tolerance through Tension-to-Thrust Feed-Forward: Hybrid Input-to-State Stability for Decentralized Multi-UAV Slung-Load Transport under Abrupt Cable Severance

Hadi Hajieghrary[1] , Paul Schmitt[2]

***Abstract*—Cooperative aerial manipulation under unannounced cable severance demands a fault response on the payload-pendulum timescale ($\approx 2\,\mathrm{s}$), well below the latency of any detection–classification–reconfiguration pipeline. For this class of self-announcing structural transitions, we show that the entire fault response collapses into a single decentralized feed-forward identity, $T_i^{\mathrm{ff}} = T_i$, equating each drone's altitude thrust feed-forward to its own measured rope tension. The identity delivers hybrid practical input-to-state stability (ISS) across unannounced severance events without any estimator, detector, or peer communication.**

We prove three closed-form guarantees. First, a taut-cable reduction theorem with $O(\delta + \eta_{\max})$ admissibility on a slack-excursion-bounded domain replaces the distributed Kelvin–Voigt rope dynamics with a lumped reduced-order plant. Second, a hybrid practical-ISS theorem yields an explicit recovery envelope with closed-loop Lyapunov decay rate λ and per-fault-cycle contraction $\rho \in (0, 1)$, both in closed form. Third, a discrete-time gain-selection bound for an optional L_1 adaptive altitude augmentation removes the dominant hyper-parameter of the architecture from empirical tuning. Because each drone's closed loop is identical pre- and post-fault and contains no peer-coupled term, the architecture admits a sequence of $F \leq N-2$ unannounced severances by induction on the fault counter, subject only to the actuator-margin envelope $m_L g/(N-F) \leq \kappa_{\mathrm{act}} f_{\max}$.

In Drake multibody simulation across a pre-declared six-variant capability sequence and four ablation campaigns at $N = 5$, $m_L = 10\,\mathrm{kg}$, and $4\,\mathrm{m\,s^{-1}}$ Dryden wind, the controller absorbs single and dual unannounced severances within one payload-pendulum period. The feed-forward ablation isolates the identity as the causal mechanism: disabling it inflates cruise RMSE by 34–39 % and post-fault altitude sag by 3.6–4.0×. Two null findings — that the model-predictive-control tension-ceiling layer and the post-fault reshape supervisor remain structurally inactive on the demonstrated trajectory family — confirm that the conventional active fault-tolerance machinery is not the mechanism doing the work.

***Index Terms*—Cooperative aerial manipulation, cable-suspended payload transport, decentralized fault-tolerant control, passive fault tolerance, hybrid practical input-to-state stability, Lyapunov contraction analysis, L_1 adaptive augmentation, receding-horizon model predictive control, anti-swing control, structural-transition systems.**

I. Introduction

Cooperative aerial transport uses N multirotor uncrewed aerial vehicles (UAVs) to lift a common payload through flexible cables. The platform has been demonstrated in laboratory and outdoor experiments for disaster-response resupply, suspended construction, and heavy-payload scenarios [1]–[4]. We focus on the regime in which fault response sets the platform's limiting behavior. Cable severance may stem from abrasion, attachment-hardware fatigue, or obstacle contact; the severed cable's tension redistributes across the remaining cables within one payload-pendulum period, and if the controller cannot absorb this redistribution within that window the payload drops. Unlike actuator or sensor faults in single-vehicle systems, cable severance is a structural change in the plant: a discrete reduction of the constraint set, not an additive perturbation. It arrives without supervisory signal and requires a corrective response on the payload-pendulum timescale, typically well under one second.

Three structural features of this plant—unilateral compliant coupling [5], hybrid constraint-cardinality transitions at severance events [6], and strict information locality (each vehicle observes only its own state, rope tension, and a local payload estimate)—rule out classical cooperative and slung-load architectures. Any admissible controller must deliver post-fault response without detection, annunciation, or supervisory reconfiguration, and must admit a closed-loop analysis that remains valid across the structural transitions.

This paper shows that, for this plant class, fault tolerance is delivered by a single feed-forward identity in the per-drone controller rather than by detection and reconfiguration: drone i's altitude thrust feed-forward is set equal to its own measured rope tension, $T_i^{\mathrm{ff}}(t) = T_i(t)$ (formally (31) in Section III).

Here T_i^{ff} is the thrust feed-forward added to drone i's commanded acceleration, and T_i is its measured rope tension. The rationale is timescale-based: when a peer's cable severs, the load redistributes across the remaining cables within one payload-pendulum period (≈ 2 s), and the surviving drone's measured tension rises during that window. Identity (31) converts that rise at the next control tick into a proportional thrust increment, absorbing the new load at the plant's native rate without detection or communication. The mechanism sits inside a conventional cascade—an outer-loop proportional-derivative (PD) controller with anti-swing slot shift, a single-step convex quadratic program (QP) on the tilt-and-thrust envelope, and a geometric attitude loop. Identity (31) is the sole nonstandard component.

We develop the consequences of (31) along three axes. The first contribution is a taut-cable reduction theorem (C1, Section II-C): within a slack-excursion-bounded domain parameterized by the pendulum-to-rope timescale ratio δ and a bound η on cable-slack duty cycle, an explicit first-order error budget justifies replacing the rope's distributed spring-damper dynamics with a lumped effective stiffness. The admissibility

[1]Hadi.Hajieghrary@Torc.ai
[2]pauls@massrobotics.org

gates (slack-run duration $\leq 40\,\mathrm{ms}$, duty cycle $\leq 2.5\%$, QP active-set-transition fraction $\leq 1\%$) are met by all six demonstrated missions with margin; worst observed values are $35.8\,\mathrm{ms}$, 1.66%, 0.10% (Table VI). The second contribution is a baseline hybrid practical input-to-state stability (ISS) theorem (C2, Section IV): on the C1 domain, under a single-active-regime doctrine for the QP active set and a spaced-fault dwell hypothesis $\tau_d \geq \tau_{\mathrm{pend}}$, the closed-loop tracking error admits a closed-form recovery envelope with Lyapunov decay rate λ and per-fault-cycle contraction $\rho \in (0, 1)$. The strict $\rho < 1$ is achieved by identity (31), which certifies Lyapunov contraction across each fault cycle rather than mere non-divergence within a pendulum period. The dwell hypothesis is structural, not conservative: it is the minimum time over which the measured-tension feed-forward $T_i^{\mathrm{ff}} = T_i$ can restore the Lyapunov function below its pre-fault level with $\rho < 1$ (Lemma 6). The third contribution is a closed-form stability bound for the L_1-adaptive altitude augmentation (C3, Section V): for the discrete-time L_1 outer loop on the altitude channel, a certifiable gain-selection condition is given in closed form in the control-tick period and the altitude-channel entry of the outer-loop Lyapunov matrix, removing the dominant hyper-parameter of L_1 from empirical tuning.

The proposed controller is validated in high-fidelity Drake multibody simulation through a pre-declared six-variant capability demonstration and four targeted stress campaigns on a five-drone, $10\,\mathrm{kg}$-payload, $4\,\mathrm{m\,s^{-1}}$ Dryden-wind scenario under single and compound cable-severance faults. The feed-forward ablation campaign isolates identity (31) as the causal mechanism: disabling it materially inflates cruise tracking error and post-fault altitude sag (Table IX). Two null findings are reported alongside: in these regimes, a composable extension layer is structurally inactive because the baseline already satisfies the relevant constraint. The architecture admits a sequence of $F \leq N - 2$ unannounced severances under the actuator-margin envelope (Section II-B); the demonstrated $N = 5$, $m_L = 10\,\mathrm{kg}$ admits $F \leq 2$.

The contribution is best read as a category claim—about how to handle structural transitions whose information arrives through actuator-side measurements on the plant's native timescale—rather than as a feature combination on a five-drone case study. Four architectural consequences follow from (31): *locality* (the admissible information set $\mathcal{I}_i$ (13) contains no peer state, no fault flag, no shared scheduler), *passivity* (the same identity that operates pre-fault produces the post-fault response, with no detector, classifier, or reassignment rule), *robustness to* $F \leq N - 2$ *unannounced severances* by induction on the fault counter under the actuator-margin envelope $m_L g/(N-F) \leq \kappa_{\mathrm{act}} f_{\max}$ (21), and *team-size independence* (the per-drone law and the lemma stack of Section IV carry no N structurally; the dependence concentrates in the actuator-margin envelope). These properties are audited against the campaign in Section VII-B; the two null findings (Remarks 5 and 6) are read here as evidence that the conventional active fault-tolerance (FT) machinery is not what is doing the work.

The remainder of the paper is organized as follows. Section II formalizes the plant, fault model, admissible controller class, reference trajectory, taut-cable reduction theorem, and the formal problem. Section III presents the decentralized controller and its three-layer cascade. Section IV establishes the main hybrid practical-ISS theorem through five supporting lemmas and characterizes its robustness. Section V adds the L_1 altitude augmentation, receding-horizon MPC with penalized tension-ceiling constraints, and the post-fault formation-reshape supervisor, and shows each preserves the baseline theorem's recovery envelope on its admissibility domain.

A. Related Work

Related work falls into four categories: centralized and distributed cooperative transport, geometric tracking and slung-load control, L_1 adaptive control, and fault-tolerant control. We map these and identify the gap this paper addresses.

a) Centralized and distributed cooperative transport.: Force-allocation architectures originating with Michael, Fink, and Kumar [1] solve a centralized QP over cable tensions to satisfy wrench-closure and actuator-limit constraints; subsequent refinements improve robustness and payload-mass estimation [2], [4] but preserve synchronous global state. Distributed predictive control [7] replaces per-tick global communication with iterative neighbor exchanges, but message complexity still scales with horizon and coupling rank. Recent work extends nonlinear MPC to six-degree-of-freedom rigid-body payload transport [8], adds control-barrier inter-vehicle safety [9], and handles parametric payload uncertainty across directed communication topologies [10]. At agile multirotor tick rates, the per-tick synchronous global-state requirement common to all of these formulations is a structural barrier to operation at rates where cable severance can occur within a single outer-loop horizon. None of these architectures accounts for cable severance in its stability argument; each assumes a fixed constraint set over the control horizon.

b) Geometric tracking and slung-load control.: The foundational single-quadrotor geometric tracking controller [11], the flatness-based slung-load framework [12], and the $\mathrm{SE}(3)$ geometric control for a rigid-body suspended payload [13] all sit under a fixed-constraint-set assumption. Multi-vehicle extensions [14], [15] reintroduce rigid-formation kinematics or global-state exchange, both of which collapse the decentralization that motivates the multi-vehicle architecture.

c) L_1 adaptive control.: L_1 adaptive control [16], [17] delivers matched-disturbance rejection with transient-performance bounds, recently extended to the full-$\mathrm{SE}(3)$ dynamics of a single quadrotor [18]; the extension to cooperative architectures under cable-fault events and strict locality remains open.

The fault-tolerant control literature divides into active architectures that reconfigure after diagnosis [19], [20] and passive architectures that absorb faults through robustness [21]. Within cooperative aerial transport, two prior strands are directly relevant. Fault-tolerant cooperative transportation under suspension-cable failures has been addressed through observer-based disturbance estimation with dynamic load reassignment [22]: each surviving vehicle estimates a tension-related disturbance, and the load allocation is redistributed by a

reassignment law. Communication-free or low-communication cooperative transport is achieved by passive force control [23] and leader–follower architectures with force estimation [24], which avoid inter-UAV message exchange but presuppose a fixed cable-cardinality regime. These strands show that neither cable-severance response nor strict locality is independently novel here. The present contribution is best read not as a feature subtraction but as a different commitment about *how the structural-transition announcement is processed*: the actuator-side tension measurement *is* the announcement, eliminating the detection-classification-reconfiguration timeline as a system-level construct rather than substituting a faster version of it. The architectural category occupied by this paper is decentralized cooperative manipulation in which discrete structural transitions self-announce through actuator-side measurements; the cable-severance instance is the certification vehicle, not the contribution.

II. Problem Statement

We study cooperative aerial transport under unannounced cable-severance faults. This section formalizes the plant, the fault model, the admissible controller class, the reference trajectory and its taut-cable reduction, and the formal problem. The regime is a continuous-time hybrid plant driven by bounded deterministic wind, subject to a finite unannounced sequence of cable-severance faults, operated under strictly local sensor access with a pre-distributed reference trajectory and per-drone actuator saturation. The remainder of the paper constructs a controller that satisfies the property-certification problem of Definition 1 and proves the guarantees summarized in Section II.

A. Plant Description and Admissible Disturbance Set

a) Vehicles, payload, and state.: A team of N quadrotor UAVs, indexed $i \in \mathcal{N} \triangleq \{1, \dots, N\}$, operates in an inertial world frame $\mathcal{W}$. Drone i has mass m_i, diagonal inertia $\boldsymbol{J}_i = \mathrm{diag}(J_x^{(i)}, J_y^{(i)}, J_z^{(i)})$, position $\boldsymbol{p}_i \in \mathbb{R}^3$, velocity $\boldsymbol{v}_i \in \mathbb{R}^3$, attitude $\boldsymbol{R}_i \in \mathrm{SO}(3)$, and angular velocity $\boldsymbol{\omega}_i \in \mathbb{R}^3$. The per-drone state is $x_i \triangleq (\boldsymbol{p}_i, \boldsymbol{v}_i, \boldsymbol{R}_i, \boldsymbol{\omega}_i)$ and the collective drone state is $x_{\mathcal{N}} \triangleq (x_i)_{i=1}^N$. A point-mass payload of mass m_L has position $\boldsymbol{p}_L \in \mathbb{R}^3$, velocity $\boldsymbol{v}_L \in \mathbb{R}^3$, and state $x_L \triangleq (\boldsymbol{p}_L, \boldsymbol{v}_L)$. We do not model payload rotation: the flexible cable attachment geometry makes rotational coupling weak relative to the translational dynamics.

b) Cables and complementarity condition.: Each drone i is connected to the payload by a flexible cable. Cable i exerts a world-frame force $\boldsymbol{T}_i^{\mathcal{W}} \in \mathbb{R}^3$ on the payload-side attachment point and an equal and opposite reaction on the drone-side attachment point. The cable transmits tensile force only when taut; its scalar tension is

$$T_i \;\triangleq\; \big\|\boldsymbol{T}_i^{\mathcal{W}}\big\| \;\geq\; 0, \tag{1}$$

with the unilateral constraint $T_i = 0$ whenever the cable is slack. The complementarity condition

$$T_i \;\geq\; 0, \quad \ell_i \;-\; L \;\leq\; 0, \quad T_i\big(\ell_i - L\big) \;=\; 0, \tag{2}$$

where $\ell_i = \|\boldsymbol{p}_{a,i} - \boldsymbol{p}_{b,i}\|$ is the cable's instantaneous length, $\boldsymbol{p}_{a,i}$ and $\boldsymbol{p}_{b,i}$ are the drone- and payload-side attachment points, and L is the rest length, captures the unilateral tensile constraint for an idealized inextensible cable. In the simulator each cable is an elastic Kelvin–Voigt spring with rest length $L = 1.25\,\mathrm{m}$ and per-segment stiffness $k_s = 25\,000\,\mathrm{N\,m^{-1}}$; $T_i > 0$ whenever $\ell_i > L$, and the elastic correction to the inextensible model is $O(\delta)$ in the timescale separation ratio δ of Section II-C. We use (2) throughout for the taut/slack distinction; the elastic correction is absorbed in the taut-cable reduction theorem. The closed-loop dynamics are non-smooth at every taut-to-slack transition.

c) Equations of motion.: The translational dynamics of drone i are governed by

$$m_i\,\dot{\boldsymbol{v}}_i \;=\; f_i\,\boldsymbol{R}_i\,\hat{\boldsymbol{e}}_3 \;-\; m_i g\,\hat{\boldsymbol{e}}_3 \;-\; \boldsymbol{T}_i^{\mathcal{W}} \;+\; \boldsymbol{F}_{\mathrm{w},i}(t), \tag{3}$$

where $f_i \in [f_{\min}, f_{\max}]$ is the scalar thrust, $\hat{\boldsymbol{e}}_3$ is the world-frame vertical unit vector, and $\boldsymbol{F}_{\mathrm{w},i}$ represents an aerodynamic wind disturbance. The rotational dynamics are

$$\boldsymbol{J}_i\,\dot{\boldsymbol{\omega}}_i + \boldsymbol{\omega}_i \times \boldsymbol{J}_i\,\boldsymbol{\omega}_i \;=\; \boldsymbol{\tau}_i, \qquad \dot{\boldsymbol{R}}_i \;=\; \boldsymbol{R}_i\,[\boldsymbol{\omega}_i]_\times, \tag{4}$$

where $\boldsymbol{\tau}_i$ is the body-frame control torque and $[\cdot]_\times$ is the skew-symmetric operator. The rope force is collapsed to an equivalent resultant acting through the drone center of mass, so cable-induced moments from attachment offsets are neglected. The payload translational dynamics are

$$m_L\,\dot{\boldsymbol{v}}_L \;=\; -m_L g\,\hat{\boldsymbol{e}}_3 + \sum_{i \in \mathcal{S}(t)} \boldsymbol{T}_i^{\mathcal{W}} + \boldsymbol{F}_{\mathrm{w},L}(t), \tag{5}$$

where $\mathcal{S}(t) \subseteq \mathcal{N}$ is the set of surviving (intact) cables at time t and $\boldsymbol{F}_{\mathrm{w},L}$ is the wind disturbance on the payload.

d) Inputs, state space, and admissible disturbance set.: The per-drone control input is

$$\begin{aligned} u_i(t) \;&\triangleq\; \big(f_i(t),\, \boldsymbol{\tau}_i(t)\big) \;\in\; \mathcal{U}_i \\ \mathcal{U}_i \;&\triangleq\; [f_{\min}, f_{\max}] \times [-\tau_{\max}, \tau_{\max}]^3, \end{aligned} \tag{6}$$

and the collective input is $u(t) \triangleq (u_i(t))_{i=1}^N$. The complete plant state is

$$\begin{aligned} x(t) \;&\triangleq\; \big(x_{\mathcal{N}}(t),\, x_L(t)\big) \;\in\; \mathcal{X} \\ \mathcal{X} \;&\triangleq\; \big(\mathbb{R}^3 \times \mathbb{R}^3 \times \mathrm{SO}(3) \times \mathbb{R}^3\big)^N \times \mathbb{R}^3 \times \mathbb{R}^3. \end{aligned} \tag{7}$$

Collecting the wind disturbances into $d(t) \triangleq (\boldsymbol{F}_{\mathrm{w},1}(t), \dots, \boldsymbol{F}_{\mathrm{w},N}(t), \boldsymbol{F}_{\mathrm{w},L}(t))$, the admissible disturbance set is

$$\begin{aligned} \mathcal{D} \;\triangleq\; \Big\{ d(\cdot) : \|\boldsymbol{F}_{\mathrm{w},i}(t)\| \leq W_{\max},\, \|\boldsymbol{F}_{\mathrm{w},L}(t)\| \leq W_{\max}, \\ \forall t \geq 0,\, \forall i \in \mathcal{N} \Big\}, \end{aligned} \tag{8}$$

where $W_{\max} > 0$ is the worst-case disturbance magnitude. No probabilistic or spectral model is assumed; the problem is posed against all bounded disturbances in $\mathcal{D}$. The Dryden realization used in V2–V6 yields per-drone peak aerodynamic force well inside the contract bound $W_{\max} = 1\,\mathrm{N}$ (drag and frontal-area parameters in Table IV).

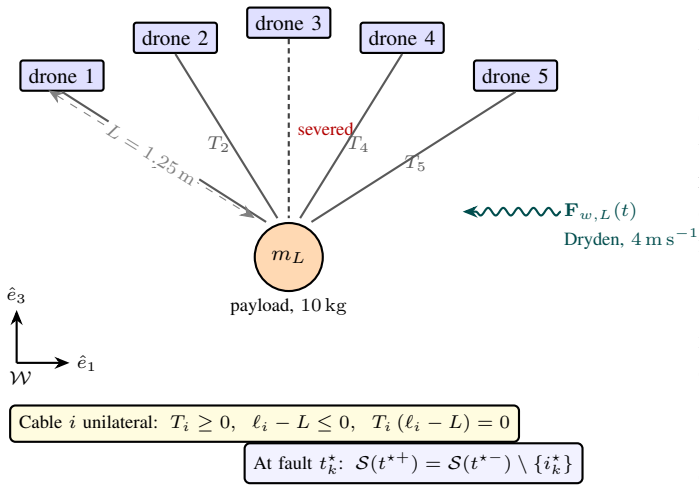


Fig. 1. Cooperative transport configuration. Five quadrotors carry a 10 kg point-mass payload through unilateral cables of rest length $L = 1.25\,\mathrm{m}$. Each drone i measures only its own state, its own scalar tension T_i, and the local payload state x_L; no peer drone or peer-cable signal is exchanged. The plant is driven by a bounded Dryden wind realisation ($4\,\mathrm{m\,s^{-1}}$ mean) and an unannounced cable-severance fault schedule $\sigma \in \mathcal{F}$ that decrements the survivor set $\mathcal{S}(t)$ at each event $t_k^\star$ (here drone $i^\star$, shown dashed).

B. Fault Model and Admissible Controller Class

a) Structural fault event.: At a time $t_k^\star$, cable $i_k^\star \in \mathcal{N}$ severs instantaneously. Severance permanently removes cable $i_k^\star$ from the constraint set:

$$T_{i_k^\star}(t) \;=\; 0 \qquad \forall\, t \geq t_k^\star. \tag{9}$$

The active cable set transitions discretely:

$$\mathcal{S}(t_k^{\star,+}) \;=\; \mathcal{S}(t_k^{\star,-}) \;\setminus\; \{i_k^\star\}, \tag{10}$$

reducing the cardinality from $|\mathcal{S}|$ to $|\mathcal{S}| - 1$. For an elastic cable, the kinematic state $(\boldsymbol{p}_i, \boldsymbol{v}_i, \boldsymbol{R}_i, \boldsymbol{\omega}_i, \boldsymbol{p}_L, \boldsymbol{v}_L)$ is continuous across severance; only the ODE right-hand side changes discontinuously at $t_k^\star$.

b) Compound faults and spaced-fault assumption.: We consider a sequence of $F \leq N-2$ fault events at times $\{t_k^\star\}_{k=1}^F$ with $t_1^\star < \cdots < t_F^\star$. The bound $F \leq N-2$ keeps at least two cables intact; actuator margins limit the demonstrated scenario ($N = 5$, $m_L = 10\,\mathrm{kg}$) to $F \leq 2$. The inter-fault interval $\tau_{d,k} \triangleq t_{k+1}^\star - t_k^\star$ is the *dwell time*. We assume each fault is separated from the previous by at least one payload-pendulum period:

$$\tau_{d,k} \;\geq\; \tau_{\mathrm{pend}} \;\triangleq\; 2\pi\sqrt{L/g}, \qquad k = 1, \ldots, F-1, \tag{11}$$

reflecting the physical requirement that load redistribution from fault k settle before fault $k+1$. At $L = 1.25\,\mathrm{m}$ and $g = 9.81\,\mathrm{m\,s^{-2}}$, $\tau_{\mathrm{pend}} \approx 2.24\,\mathrm{s}$. The admissible fault-schedule set is

$$\mathcal{F} \;\triangleq\; \Big\{\{(t_k^\star, i_k^\star)\}_{k=1}^F : F \leq N-2, \tag{12}$$
$$t_{k+1}^\star - t_k^\star \geq \tau_{\mathrm{pend}},\ i_k^\star \in \mathcal{S}(t_k^{\star,-})\Big\},$$

$F = 0$ corresponds to the no-fault case.

c) Information unavailability.: No fault flag, annunciation, or detection signal is transmitted. Each surviving drone $i \in \mathcal{S}(t)$ continues executing its nominal controller without explicit knowledge of the fault. The fault manifests implicitly through two local signals in $\mathcal{I}_i(t)$: a rise in the drone's own rope tension $T_i(t)$ as the load redistributes, and a transient deviation of the payload velocity $\boldsymbol{v}_L(t)$ from reference. No fault-detection logic is required; the controller responds passively to these signals.

d) Admissible controller class.: Field-deployable avionics impose a strict *admissible information pattern*. Drone i's controller π_i is restricted to the information set

$$\mathcal{I}_i(t) \;\triangleq\; \big\{x_i(t),\ T_i(t),\ \boldsymbol{v}_L(t),\ \boldsymbol{p}_L^d(\cdot),\ \boldsymbol{v}_L^d(\cdot)\big\}, \tag{13}$$

comprising the drone's inertial state x_i, its rope-tension measurement T_i, the payload velocity $\boldsymbol{v}_L$ (measured via a co-located downward-facing optical-flow sensor, no drone-to-drone communication), and a pre-distributed reference $(\boldsymbol{p}_L^d, \boldsymbol{v}_L^d)$ loaded before flight. Payload position $\boldsymbol{p}_L$ is deliberately excluded; only velocity enters the closed loop, via the anti-swing slot shift of Section III. Each π_i is a causal feedback map,

$$u_i(t) \;=\; \pi_i\Big(\mathcal{I}_i|_{[0,t]}\Big) \in \mathcal{U}_i, \tag{14}$$

possibly with a finite-dimensional internal state, but never with access to information outside (13). The admissible controller class is denoted Π_{loc}. Explicitly, $\mathcal{I}_i(t)$ excludes any peer-drone state x_j, any peer-cable tension T_j, any fault flag, detection signal, or supervisory message, and any run-time inter-drone packet. A controller satisfying this restriction is *decentralized* and *strictly local*: its output uses only co-located sensors and a pre-flight plan.

A structural fault—a discrete change in constraint cardinality—is *self-announcing* with respect to $\mathcal{I}_i(t)$ when the change produces, within the plant's native response timescale, a directly measured local signal $s(t) \in \mathcal{I}_i(t)$ that the nominal feedback law converts into the required compensatory action without detection, classification, or reconfiguration. Cable severance is the canonical instance: the surviving tension $T_i(t)$ rises within τ_{rope} of each severance, and the identity $T_i^{\mathrm{ff}} = T_i$ converts that rise into a thrust increment at the next control tick.

C. Reference Trajectory and Taut-Cable Reduction

a) Reference trajectory.: The desired payload motion is a three-dimensional lemniscate trajectory with a sinusoidal altitude variation,

$$\boldsymbol{p}_L^d(t) \;=\; \begin{bmatrix} \dfrac{a\cos\phi(t)}{1+\sin^2\phi(t)} \\ \dfrac{a\sin\phi(t)\cos\phi(t)}{1+\sin^2\phi(t)} \\ z_0 + h_z\sin(\phi(t)) \end{bmatrix}, \tag{15}$$
$$\phi(t) = \frac{2\pi(t-t_0)}{T_{\mathrm{ref}}},$$

where $a = 3.0\,\mathrm{m}$ is the horizontal scale, $T_{\mathrm{ref}} = 12\,\mathrm{s}$ the period, $z_0 = 3.0\,\mathrm{m}$ the cruise altitude, and $h_z = 0.35\,\mathrm{m}$ the altitude

excursion amplitude. These parameters yield a peak centripetal load of $\approx 0.25g$ at the figure-8 tips. The analytical curve (15) is $\mathcal{C}^\infty$; in the simulator it is pre-sampled at $\Delta t_{\mathrm{ref}} = 0.2\,\mathrm{s}$ and linearly interpolated, with the discretization residual absorbed into the formal problem's practical tracking bounds.

b) Rope model and lumped effective stiffness.: The stability analysis of Section IV operates on a finite-dimensional reduced-order model rather than the full distributed Kelvin–Voigt rope dynamics. We establish that the reduction is valid on a slack-excursion-bounded domain, with an explicit first-order error bound. Each rope of rest length L is discretized into $N_b = 8$ identical Kelvin–Voigt beads of mass m_{bead} connected by $N_{\mathrm{seg}} = N_b + 1 = 9$ segments of stiffness k_s and damping c_s; N_{seg} is reserved for rope segments and is distinct from the surviving-cable cardinality $N_s(t) = |\mathcal{S}(t)| = N - F(t)$ used in Section IV. The distributed dynamics evolve on a fast timescale

$$\tau_{\mathrm{rope}} \;=\; 2\pi\sqrt{m_{\mathrm{bead}}/k_s} \;\approx\; 6.3\,\mathrm{ms} \tag{16}$$

separated from the payload-pendulum timescale $\tau_{\mathrm{pend}} \approx 2.24\,\mathrm{s}$ by $\delta = \tau_{\mathrm{rope}}/\tau_{\mathrm{pend}} \approx 0.003$. On the slow manifold, the net tensile force at the drone-side attachment equals the scalar $\xi_i = k_{\mathrm{eff}}(\ell_i - L)^+$, with effective stiffness

$$k_{\mathrm{eff}} \;=\; \frac{k_s}{N_{\mathrm{seg}}} \;=\; \frac{25\,000\,\mathrm{N\,m^{-1}}}{9} \;\approx\; 2778\,\mathrm{N\,m^{-1}}. \tag{17}$$

The six pre-registered missions (V1–V6, Section VI-A) keep the system inside $\Omega_\tau^{\mathrm{dwell}}$ with margin (worst slack run $35.8\,\mathrm{ms}$, worst duty cycle 1.66%, Table VI). The reduction replaces the $3N_b$-dimensional bead-chain state with the scalar ξ_i; the per-instant taut error is $O(\delta)$ and slack excursions add $O(\eta_{\max})$.

Theorem 1 (Taut-cable reduction)**.** *On the slack-excursion-bounded domain*

$$\Omega_\tau^{\mathrm{dwell}} \;\triangleq\; \big\{x \in \mathcal{X} : \textit{every slack run} \le \tau_{\mathrm{slack,max}},\ \textit{duty cycle} \le \eta_{\max}\big\}, \tag{18}$$

with $\tau_{\mathrm{slack,max}} = 40\,\mathrm{ms}$ and $\eta_{\max} = 2.5\%$, the distributed Kelvin–Voigt rope dynamics are approximated by the lumped scalar model $T_i = k_{\mathrm{eff}}(\ell_i - L)^+$ with closed-loop error

$$\|T_i^{\mathrm{true}} - T_i^{\mathrm{lumped}}\| \;\le\; C_1\,\delta + C_2\,\eta_{\max}, \tag{19}$$

for constants $C_1, C_2 > 0$ depending on the bead-chain parameters $(m_{\mathrm{bead}}, c_s, k_s, N_b)$ and the wind bound $W_{\max}$. The reduced-order model is therefore admissible as the plant for the stability analysis of Section IV.

Sketch. The bead-chain spectrum is separated from the pendulum by $\delta \approx 0.003$, so a standard singular-perturbation argument applies on any taut interval. Tikhonov's theorem then gives $\|T_i^{\mathrm{bead}} - T_i^{\mathrm{lumped}}\| = O(\delta)$ uniformly in $W_{\max}$, with C_1 set by bead mass, damping, stiffness, and maximum external load. Each slack run of duration τ_{s} has amplitude $O(W_{\max}\tau_{\mathrm{s}}^2/m_L)$ and is followed by a re-tension transient closing within one rope period, leaving a residual of $O(\eta_{\max})$ once integrated against the duty-cycle budget; C_2 is proportional to $W_{\max}$ and to the re-tension amplitude and is independent of $|\mathcal{S}(t)|$. Both constants are finite on $\Omega_\tau^{\mathrm{dwell}}$; their numerical values are not needed, since the stability analysis absorbs the $O(\delta + \eta_{\max})$ residual into the class-$\mathcal{K}$ gain γ_η of Theorem 7. □

c) Empirical domain membership.: Table VI reports slack-excursion statistics for all six missions. Worst-case slack-run duration is $35.8\,\mathrm{ms}$ (V2–V5, $4.2\,\mathrm{ms}$ margin); worst-case slack duty cycle is 1.66% (V6, 0.85 pp margin); worst-case QP active-set-transition fraction is 0.10% against the D1 threshold of 1%. All six missions reside inside $\Omega_\tau^{\mathrm{dwell}}$ with non-trivial margin on every gate. The H1 gate values are physically motivated, not data-fitted: $\tau_{\mathrm{slack,max}} = 40\,\mathrm{ms}$ is one quarter of the slow-pendulum frozen window $\tau_{\mathrm{pend}}/(2\pi)$, the duration over which a single slack excursion does not propagate dispersively along the bead chain; $\eta_{\max} = 2.5\%$ is set against the Rayleigh statistics of the turbulent-crosswind response so that slack events arrive at a rate the re-tension cycle can absorb. The theorem is therefore asserted on the regime occupied by the demonstrated missions, not on an untested all-taut idealisation.

d) Team-size independence and actuator-margin envelope.: The per-drone controller and the lemma stack of Section IV carry no team size N structurally. The dependence enters through two scalars: per-drone hover load $T^{\mathrm{hov}}(N) = m_L g/N$ and post-fault redistributed load on the surviving set,

$$T^{\mathrm{post}}(N, F) \;=\; \frac{m_L g}{N - F} \;=\; T^{\mathrm{hov}}(N) \cdot \frac{N}{N - F}, \tag{20}$$

where F is the number of severed cables. The surcharge factor $N/(N-F)$ inflates per-drone tension by $1/(N-F)$ relative to the no-fault baseline and is the only N-dependent quantity in the recovery envelope of Theorem 7.

The architecture is therefore designable across N by a single *actuator-margin condition*: a configuration $(N, m_L, F, f_{\max})$ is admissible if

$$\frac{m_L\, g}{N - F} \;\le\; \kappa_{\mathrm{act}}\, f_{\max}, \tag{21}$$

where $\kappa_{\mathrm{act}} \in (0, 1)$ reserves headroom for tilt projection, anti-swing residual, and dynamic centripetal load. With $\kappa_{\mathrm{act}} = 0.82$, $f_{\max} = 150\,\mathrm{N}$, and the demonstrated $(N, m_L, F) = (5, 10\,\mathrm{kg}, 2)$, the inequality reads $32.7\,\mathrm{N} \le 123\,\mathrm{N}$, i.e. 27% of the actuator envelope, leaving designable extension along (N, m_L, F). Figure 2 plots the boundary curves for $F \in \{0, 1, 2, 3\}$ over $N \in \{3, \ldots, 10\}$ and marks the demonstrated point. The $N = 5$ scenario is one point on a designable feasibility surface; the certificate of Section IV carries unchanged at any (N, m_L, F) satisfying (21).

With the plant, fault model, reference trajectory, and reduction in place, we state the formal problem. Unilateral cable coupling (2), discrete structural fault transitions (9)–(10), admissible disturbances (8), and strict locality (13) together yield a hybrid decentralized control problem outside standard cooperative-transport formulations. The design variable is not a centralized control sequence but a family of local causal feedback laws. The problem, posed on the reduced-order model certified by Theorem 1, is as follows.

Definition 1 (Fault-tolerant decentralized aerial transport)**.** *Given the plant* (3)–(5)*, the complementarity constraint* (2)*,*

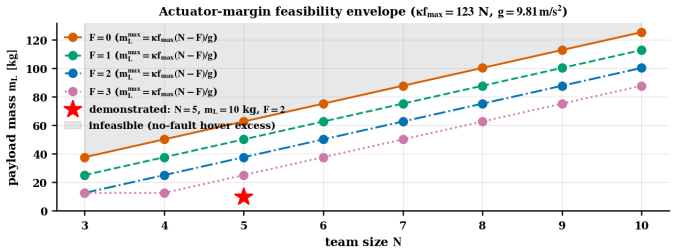


Fig. 2. Actuator-margin feasibility envelope (21) in (N, m_L) space at fault counts $F \in \{0,1,2,3\}$, $\kappa_{\mathrm{act}} f_{\max} = 123\,\mathrm{N}$. Red star: demonstrated (N=5, m_L=10 kg, F=2).

the mission reference (15)*, all admissible disturbances* $d(\cdot) \in \mathcal{D}$ (8)*, and all admissible fault schedules* $\sigma \in \mathcal{F}$ (12)*, find a causal controller family* $\{\pi_i\}_{i=1}^N \in \Pi_{\mathrm{loc}}^N$ *such that the closed-loop system satisfies the properties* ***P1–P3*** *for every* $d(\cdot) \in \mathcal{D}$ *and every* $\sigma \in \mathcal{F}$*. This is a property-certification problem rather than a scalar optimal-control problem: the design succeeds if and only if* ***P1–P3*** *hold jointly.*

P1 ***(Nominal tracking).*** *In the absence of faults and for all admissible disturbances* $d(\cdot) \in \mathcal{D}$*, the payload tracking error* $e_L(t) \triangleq \boldsymbol{p}_L(t) - \boldsymbol{p}_L^d(t)$ *is ultimately practically bounded:*

$$\limsup_{t\to\infty} \|e_L(t)\| \;\le\; \bar{e}_\infty + \bar{e}_0(W_{\max}), \tag{22}$$

where $\bar{e}_\infty \ge 0$ *captures non-vanishing residuals from finite-gain tracking, cable-elastic reduction, and reference discretization, and* $\bar{e}_0 : \mathbb{R}_{\ge 0} \to \mathbb{R}_{\ge 0}$ *is class-*$\mathcal{K}$ *with* $\bar{e}_0(0) = 0$*.*

P2 ***(Passive fault absorption).*** *After any fault* $t_k^\star$ *satisfying* (11)*, the payload tracking error recovers to a bounded envelope without detection, annunciation, or reconfiguration: there exist constants* $\Gamma_f, \lambda > 0$ *and a per-fault-cycle contraction* $\rho \in (0,1)$ *such that*

$$\begin{aligned}\|e_L(t)\| \;\le\; & \beta\big(\|e_L(t_k^{\star,-})\|,\, t - t_k^\star\big) \\ & + \Gamma_f\, e^{-\lambda(t-t_k^\star)} \\ & + \bar{e}_\infty + \bar{e}_0(W_{\max}), \quad t \ge t_k^\star,\end{aligned} \tag{23}$$

for some class-$\mathcal{KL}$ *function* β*. The contraction* $\rho \in (0,1)$ *certifies bounded Lyapunov growth across successive faults:* $V(x(t_{k+1}^{\star,-})) \le \rho V(x(t_k^{\star,-})) + c$ *for an explicit* $c > 0$*. The tracking error need not decrease monotonically at individual events (the payload droops and recovers), but the fault-to-fault envelope is bounded.*

P3 ***(Actuator feasibility).*** *For all* $t \ge 0$ *and all admissible fault sequences, the thrust* $f_i(t) \in [f_{\min}, f_{\max}]$ *and torque* $\boldsymbol{\tau}_i(t) \in [-\tau_{\max}, \tau_{\max}]^3$ *remain inside the actuator envelope with non-zero margin.*

e) Why the problem is hard.: Three structural obstacles block classical solutions. *(i)* The unilateral cable constraint (2) makes the closed-loop vector field non-smooth at every taut-to-slack transition, so analysis must be valid on the slack-bounded set $\Omega_\tau^{\mathrm{dwell}}$ rather than a globally Lipschitz domain. *(ii)* A cable severance is a discrete decrement of the constraint set, not an additive perturbation: $|\mathcal{S}|$ drops by one and the equilibrium tension distribution shifts instantaneously, requiring a hybrid Lyapunov argument across finitely many such jumps. We formalize this through the tension-error state

$$\boldsymbol{\varepsilon}_T(t) \;\triangleq\; \boldsymbol{T}(t) \;-\; \boldsymbol{T}^{\mathrm{qs}}(t), \tag{24}$$

where $\boldsymbol{T}^{\mathrm{qs}}(t)$ is the slow-manifold tension profile restricted to the active cable set $\mathcal{S}(t)$. *(iii)* Strict information locality (13) blocks centralized fault-detection and peer-state-aggregation architectures; the controller must absorb the post-fault load increment using only the local rise in $T_i(t)$. No existing cooperative aerial transport formulation addresses these three jointly.

f) Forward references.: The reduction theorem (Theorem 1) gives the $O(\delta + \eta_{\max})$ admissibility bound on $\Omega_\tau^{\mathrm{dwell}}$; the hybrid practical-ISS envelope with closed-form λ and $\rho \in (0,1)$ is Theorem 7 of Section IV. The closed-form L_1 gain bound is Proposition 1 in Section V. Acceptance thresholds and performance targets are in Section VI.

III. Proposed Method

We present the decentralized fault-tolerant controller that solves Definition 1 on the reduced-order model of Section II-C. Each drone i runs one instance of the *DecentralizedLocalController*, a continuous-time Drake `LeafSystem<double>`. The controller is a three-layer cascade: an outer-loop PD law with a passive anti-swing slot shift, a single-step convex QP that enforces the tilt-and-thrust actuator envelope while absorbing the feed-forward tension, and an inner-loop Euler-angle attitude PD. The central design element is $T_i^{\mathrm{ff}} = T_i$: setting the thrust feed-forward equal to the drone's measured rope tension encodes passive fault tolerance with no detection, communication, or supervisory reconfiguration. This is a property of *the baseline*; the optional reshape supervisor of Section V-C adds a local tension-latch detector and a per-fault broadcast and therefore lies outside the baseline information pattern, as we make explicit there. The optional L_1-adaptive altitude augmentation is detailed in Section V as a discrete sub-system updated at fixed period $\Delta t_{L_1} \equiv T_S = 0.2\,\mathrm{ms}$ (T_s used in the stability analysis of Section V-A); its injection point is identified in Section III-B.

A. Controller Input–Output Signature

At each continuous-time evaluation, drone i's controller consumes its own inertial state $(\boldsymbol{p}_i, \boldsymbol{v}_i, \boldsymbol{R}_i, \boldsymbol{\omega}_i^{\mathcal{W}})$, scalar rope tension $T_i(t) \ge 0$, locally observed payload velocity $\boldsymbol{v}_L(t)$, the pre-distributed reference $(\boldsymbol{p}_L^d(t), \boldsymbol{v}_L^d(t))$ linearly interpolated between samples, and optionally a slot-offset override $\boldsymbol{\delta}_i^{\mathrm{ovr}}(t)$ from a *FormationCoordinator* port. When the port is disconnected, the static formation offset $\boldsymbol{\delta}_i$ from the parameter struct is used. The baseline controller does not access payload position $\boldsymbol{p}_L$. The information consumed is a strict subset of $\mathcal{I}_i(t)$ (13); no peer state, peer-cable tension, or fault signal is accessed. The controller emits the world-frame wrench $(\boldsymbol{F}_i^{\mathcal{W}}, \boldsymbol{\tau}_i^{\mathcal{W}})$ applied to the drone body.

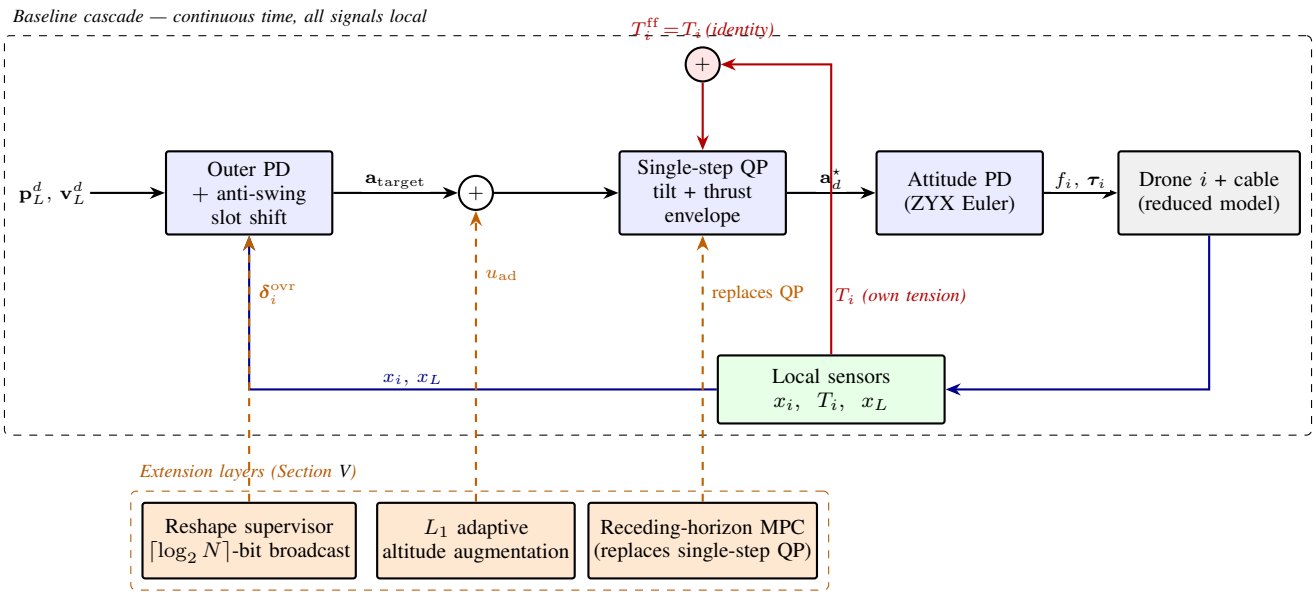


Fig. 3. Per-drone control cascade. The baseline (blue, framed top) is a three-stage continuous-time pipeline: outer-loop PD with anti-swing slot shift, a single-step convex QP that projects $\mathbf{a}_{\mathrm{target}}$ onto the tilt-and-thrust envelope, and an Euler-angle attitude PD. The non-standard element is the red feed-forward tap: drone i routes its own measured tension T_i directly into the altitude thrust feed-forward, $T_i^{\mathrm{ff}} = T_i$. This single identity is the carrier of hybrid practical-ISS across cable-severance events (Theorem 7); the surrounding cascade is otherwise textbook. Three optional layers (orange, framed bottom) hook in at three precisely identified injection points: the L_1 adaptive correction u_{ad} enters the altitude channel between the outer PD and the QP; the receding-horizon MPC replaces the single-step QP entirely; the reshape supervisor overrides the formation slot $\boldsymbol{\delta}_i$ on receipt of a single $\lceil \log_2 N \rceil$-bit broadcast.

TABLE I
COMPONENT-LEVEL INFORMATION PATTERN. THE BASELINE CASCADE AND THE L_1 AND MPC EXTENSIONS CONSUME NO FAULT FLAG AND NO PEER MESSAGE; ONLY THE OPTIONAL RESHAPE SUPERVISOR STEPS OUTSIDE THIS PATTERN.

Component	Detection?	Communication?	In main theorem?	Purpose
Baseline cascade + identity (31)	No	No	Yes	Passive fault absorption
L_1 altitude augmentation	No	No	Preserves theorem	Parameter mismatch
MPC tension ceiling	No	No	Preserves theorem	Tension constraint
Reshape supervisor	Yes (latch)	Yes ($\lceil \log_2 N \rceil$ bits)	Optional only	Slot reassignment

B. Outer Loop: PD with Anti-Swing Slot Shift

a) Slot reference and anti-swing shift.: Each drone is pre-assigned a hovering slot displaced vertically above the payload reference point and laterally by a fixed formation offset,

$$\boldsymbol{p}_{\mathrm{slot},i}(t) \;=\; \boldsymbol{p}_L^d(t) + \boldsymbol{\delta}_i + d_{\mathrm{rope}}\,\hat{\boldsymbol{e}}_3, \tag{25}$$

where $\boldsymbol{\delta}_i \in \mathbb{R}^3$ is the formation offset (overridable at run time), $d_{\mathrm{rope}} = 1.25\,\mathrm{m}$ is the vertical slot elevation matching L, and $\hat{\boldsymbol{e}}_3$ is the upward unit vector. The slot velocity is $\boldsymbol{v}_{\mathrm{slot},i}(t) = \boldsymbol{v}_L^d(t)$. To damp the payload pendulum passively, we shift the slot relative to the payload's horizontal velocity. With $\boldsymbol{v}_{L,\perp} \triangleq [v_{L,x}, v_{L,y}, 0]^\top$, the horizontal swing correction is

$$\boldsymbol{s}_i(t) = \begin{cases} -k_{\mathrm{swing}}\,\boldsymbol{v}_{L,\perp} & \text{if } k_{\mathrm{swing}}\|\boldsymbol{v}_{L,\perp}\| \le s_{\max}, \\ -s_{\max}\,\hat{\boldsymbol{v}}_{L,\perp} & \text{otherwise}, \end{cases} \tag{26}$$

where $\hat{\boldsymbol{v}}_{L,\perp}$ is the unit vector of $\boldsymbol{v}_{L,\perp}$. The shifted slot position is $\boldsymbol{p}_{\mathrm{slot},i}^{\mathrm{dyn}}(t) = \boldsymbol{p}_{\mathrm{slot},i}(t) + \boldsymbol{s}_i(t)$.

b) Tracking errors.: The position and velocity errors are

$$\boldsymbol{e}_p(t) = \boldsymbol{p}_{\mathrm{slot},i}^{\mathrm{dyn}}(t) - \boldsymbol{p}_i(t), \qquad \boldsymbol{e}_v(t) = \boldsymbol{v}_{\mathrm{slot},i}(t) - \boldsymbol{v}_i(t). \tag{27}$$

The velocity error uses the unshifted slot velocity; the anti-swing correction modifies the position reference only.

c) Target acceleration.: The outer loop computes a target body acceleration,

$$\boldsymbol{a}_{\mathrm{target}}(t) = \underbrace{\begin{bmatrix} K_p^{xy}\,e_{p,x} + K_d^{xy}\,e_{v,x} \\ K_p^{xy}\,e_{p,y} + K_d^{xy}\,e_{v,y} \\ K_p^{z}\;\,e_{p,z} + K_d^{z}\;\,e_{v,z} \end{bmatrix}}_{\boldsymbol{a}_{\mathrm{track}}} + \; w_{\mathrm{swing}} \begin{bmatrix} -k_{\mathrm{swing}}\,v_{L,x} \\ -k_{\mathrm{swing}}\,v_{L,y} \\ 0 \end{bmatrix}. \tag{28}$$

The altitude gain $K_p^z = 100$ exceeds the horizontal gain $K_p^{xy} = 30$ because vertical rope forces dominate the disturbance budget; the formal pole and Lyapunov-matrix analysis is in Theorem 3. When L_1 is active, its altitude correction $u_{\mathrm{ad}}(t)$ is added to the z-component of $\boldsymbol{a}_{\mathrm{target}}$ between the outer PD and the QP (Section V); horizontal channels are unmodified.

C. Actuator Projection and Feed-Forward Tension

a) Single-step convex QP.: $\boldsymbol{a}_{\mathrm{target}}$ may lie outside the tilt-and-thrust actuator envelope. We project onto the feasible

set with a single-step convex QP that also regularises large commands:

$$\boldsymbol{a}_d^\star = \underset{\boldsymbol{a}_d \in \mathbb{R}^3}{\operatorname{argmin}}\ w_t \left\|\boldsymbol{a}_d - \boldsymbol{a}_{\text{target}}\right\|^2 + w_e \left\|\boldsymbol{a}_d\right\|^2, \tag{29a}$$

$$\text{s.t.}\ \ |a_{d,x}|,\ |a_{d,y}| \le a_{\text{tilt}} \triangleq g \tan\theta_{\max}, \tag{29b}$$

$$a_{z,\min}(t) \le a_{d,z} \le a_{z,\max}(t), \tag{29c}$$

where the vertical bounds incorporate the rope tension feed-forward,

$$a_{z,\min}(t) = \frac{f_{\min} - T_i^{\text{ff}}(t)}{m_i} - g, \tag{30}$$

$$a_{z,\max}(t) = \frac{f_{\max} - T_i^{\text{ff}}(t)}{m_i} - g.$$

At canonical parameters, $a_{\text{tilt}} = 9.81 \tan(0.6) \approx 6.71\,\text{m}\,\text{s}^{-2}$.

b) Feasibility.: The vertical interval in (29c) is non-empty whenever $f_{\min} < f_{\max}$, since both bounds shift by the same feed-forward term, and hover feasibility holds in all scenarios ($T_i \ll f_{\max}$). Doctrine D1 (single-active-regime QP, formalised as H3 in Section IV-A) is empirically satisfied at the 0.10% worst transition fraction in Table VI; the component-wise-clamp fallback is implemented but has not been triggered.

c) Feed-forward tension.: For the canonical campaigns `initial_pretensioned=true`, so the feed-forward is the identity from $t = 0$:

$$\boxed{T_i^{\text{ff}}(t) \ = \ T_i(t).} \tag{31}$$

A smoothstep pickup ramp $\alpha(\tau) = 3\tau^2 - 2\tau^3$ handles cold-start engagement; it is bypassed in every reported campaign. The flag `disable_tension_ff` overrides (31) to $T_i^{\text{ff}} = 0$ for the feed-forward ablation campaign.

Remark 1 (Passive fault tolerance via (31))**.** *Identity* (31) *is necessary, not merely convenient, for the dwell-cycle contraction $\rho \in (0,1)$ of Theorem 6. The bounded fault-jump of Theorem 5 keeps the post-fault overshoot of leading order in the redistribution magnitude Δ_f: drone i's thrust tracks the new measured tension within the rope timescale, so the altitude-channel contribution stays of order Δ_f rather than inflating by an uncompensated matched disturbance. Without* (31)*, the altitude channel sees the redistribution as an unrejected matched disturbance, and the overshoot exceeds what the dwell-cycle decay can absorb within one pendulum period. The ablation campaign confirms this: disabling* (31) *raises post-fault RMS tracking error by* 34–39% *and inflates the altitude-sag envelope by* 3.6–4.0× *across V3, V4, V5.*

D. Inner Loop: Attitude PD and Controller Parameters

a) Thrust and desired attitude.: Given the QP solution $a_{d,z}^\star$, the commanded thrust is

$$f_i \ = \ \text{clamp}\big(m_i(g + a_{d,z}^\star) + T_i^{\text{ff}},\ f_{\min},\ f_{\max}\big). \tag{32}$$

Under the small-angle approximation, the horizontal acceleration commands map to desired pitch and roll angles,

$$\theta_{\text{pitch}}^d = \text{sat}\left(\frac{a_{d,x}^\star}{g},\ \pm\theta_{\max}\right), \tag{33}$$

$$\theta_{\text{roll}}^d = \text{sat}\left(-\frac{a_{d,y}^\star}{g},\ \pm\theta_{\max}\right).$$

TABLE II
PROPOSED CONTROLLER PARAMETERS (CANONICAL VALUES USED IN ALL SIMULATION CAMPAIGNS).

Parameter	Value	Role
$K_p^{xy},\ K_d^{xy}$	30, 15	horizontal outer loop
$K_p^z,\ K_d^z$	100, 24	altitude outer loop
$K_p^{\text{att}},\ K_d^{\text{att}}$	25, 4	attitude inner loop
k_{swing}	0.8	anti-swing gain
w_{swing}	0.3	anti-swing weight in $\boldsymbol{a}_{\text{target}}$
$s_{\max}$	0.3 m	slot-shift saturation
d_{rope}	1.25 m	slot z-offset (nominal rope length)
$w_t,\ w_e$	1.0, 0.02	QP tracking / effort weights
$\theta_{\max}$	0.6 rad	tilt ceiling (≈34°)
$f_{\min},\ f_{\max}$	0, 150 N	thrust envelope
$\tau_{\max}$	10 N m	torque saturation
T_{thresh}	0.3 N	pickup-detection threshold
$\bar{T}$	19.62 N	nominal per-drone tension ($m_L g/N$)
$K_{p,T}$	1.5	pickup-phase tension correction gain
T_{ramp}	1.0 s	pickup smoothstep window
Δt_{L_1}	0.2 ms	L_1 discrete update period

b) Euler-angle attitude PD and output wrench.: Roll and pitch are extracted from $\boldsymbol{R}_i$ by the standard ZYX decomposition; yaw uses the skew component $\epsilon_\psi = \frac{1}{2}(R_{21} - R_{12})$. With body angular velocity $\boldsymbol{\omega}_i^{\mathcal{B}} = \boldsymbol{R}_i^\top \boldsymbol{\omega}_i^{\mathcal{W}}$, the body torques are

$$\tau_{i,x} = K_p^{\text{att}}\big(\theta_{\text{roll}}^d - \phi_{\text{roll}}\big) - K_d^{\text{att}}\,\omega_{i,x}^{\mathcal{B}}, \tag{34}$$

$$\tau_{i,y} = K_p^{\text{att}}\big(\theta_{\text{pitch}}^d - \theta_{\text{pitch}}\big) - K_d^{\text{att}}\,\omega_{i,y}^{\mathcal{B}}, \tag{35}$$

$$\tau_{i,z} = -\frac{K_p^{\text{att}}}{2}\,(R_{21} - R_{12}) - K_d^{\text{att}}\,\omega_{i,z}^{\mathcal{B}}, \tag{36}$$

each saturated to $[-\tau_{\max}, \tau_{\max}]$. The law (34)–(36) uses Euler-angle error directly rather than a geodesic $\text{SO}(3)$ map; this is valid in the small-tilt regime enforced by (29b). The output wrench is

$$\boldsymbol{F}_i^{\mathcal{W}} = f_i\,\boldsymbol{R}_i\,\hat{\boldsymbol{e}}_3, \qquad \boldsymbol{\tau}_i^{\mathcal{W}} = \boldsymbol{R}_i\,[\tau_{i,x},\ \tau_{i,y},\ \tau_{i,z}]^\top, \tag{37}$$

applied at the same continuous-time evaluation instant as the upstream computations.

c) Canonical controller parameters.: Table II collects the parameters used throughout all simulation campaigns.

The three-layer controller specified above is the baseline on which Section IV establishes the hybrid practical-ISS theorem. The only structurally nonstandard element is the feed-forward identity $T_i^{\text{ff}} = T_i$ in (30) and (32); every other block is conventional. The next section shows that this single identity is the mechanism responsible for the per-fault-cycle Lyapunov contraction, with $\rho \in (0,1)$, and that without it, the closed loop loses its hybrid practical-ISS property.

IV. Stability Analysis

The baseline controller of Section III achieves fault tolerance through a single structural choice: setting the altitude thrust feed-forward equal to the drone's own measured rope tension, $T_i^{\text{ff}} = T_i$. We establish that this identity renders the closed loop *hybrid practical input-to-state stable* (hybrid practical-ISS) under the cable-severance fault model of Section II-C. When a peer cable severs, T_i rises immediately

and the feed-forward converts that step into a proportional thrust increment at the plant's native rate, without any detection, annunciation, or reconfiguration interval. The argument proceeds through the analytical hypotheses, five lemmas, the main theorem, and a claims-to-evidence map.

a) Status of the lemma stack.: The five lemmas and the main theorem certify the architectural claim of Section I—that for cooperative manipulation under self-announcing structural transitions, a single decentralized feed-forward identity suffices—rather than constituting a stand-alone theoretical contribution. The hybrid-systems framework is borrowed from [6]; the practical-ISS composition tools are standard. The value added is showing that the framework's hypotheses hold for the proposed architecture on the declared admissibility domain $\Omega_\tau^{\mathrm{dwell}}$, with closed-form expressions for ρ, λ, and the per-fault jump $\chi(\Delta_f)$.

A. Setup, Notation, and Analytical Hypotheses

The analysis operates on the reduced-order model of Section II-C. Let $e_{p,z,i}(t) = p_{\mathrm{slot},i,z}(t) - p_{i,z}(t)$ be drone i's altitude slot-tracking error and $\boldsymbol{e}_p(t) \in \mathbb{R}^{3N_s}$ the stacked position error over $\mathcal{S}(t)$ with $N_s = |\mathcal{S}(t)|$. Fault events and the dwell condition follow Section II-B; consecutive faults are separated by at least one payload-pendulum period,

$$\tau_{d,k} \;\geq\; \tau_{\mathrm{pend}} \;\triangleq\; 2\pi\sqrt{L/g}. \tag{38}$$

At $L = 1.25\,\mathrm{m}$, $\tau_{\mathrm{pend}} = 2.24\,\mathrm{s}$ and the pendulum natural frequency is $\omega_p \triangleq \sqrt{g/L} = 2.80\,\mathrm{rad\,s^{-1}}$.

The analysis rests on three hypotheses, defended empirically in Tables III and VI. **H1** (slack-excursion budget) requires every trajectory to remain in $\Omega_\tau^{\mathrm{dwell}}$: maximum slack-run duration $\tau_{\mathrm{slack,max}} \leq 40\,\mathrm{ms}$ and duty cycle $\eta_{\max} \leq 2.5\%$. **H2** (spaced-fault dwell) requires $\tau_{d,k} \geq \tau_{\mathrm{pend}}$ (38). **H3** (Doctrine D1, single-active-regime QP) requires the per-tick QP's active set $\mathcal{A}(t)$ to be locally constant inside each inter-fault window; transitions are absorbed into the H1 $\eta_{\max}$ budget and empirically hold on $\geq 99.9\%$ of mission time (worst transition fraction 0.10%, Table VI). Claims outside H1–H3—rapid multi-fault, long slack excursions, QP-fallback—are out of scope; the analysis produces no bounds for trajectories violating them.

The empirical defence of H1 and H3 uses the same pre-declared missions as the theorem's consequences, so the validation is partially *post hoc*: the bounds on $\tau_{\mathrm{slack,max}}$, $\eta_{\max}$, and the QP transition fraction are observed rather than guaranteed by design. The theorem therefore provides a conditional guarantee, and the margins of Table VI document how far inside the domain the missions sit. Design-time certification of H1 and H3 for unseen trajectory families requires trajectory-planning machinery beyond the present scope and is identified as future work. The reduction theorem (Theorem 1, C1) is independently validated against the bead-chain truth model in Section VI-B, providing an *a priori* check that the reduced model tracks the full plant within $O(\delta+\eta_{\max})$; this separation addresses the main circularity concern.

B. Lemmas Underpinning the Hybrid Practical-ISS Theorem

We state five lemmas that support the main theorem. Theorem 2 establishes algebraic tension cancellation on the altitude channel under a cable-tilt domain condition; Theorem 3 gives exponential ISS through the closed-loop Lyapunov matrix; Theorem 4 shows the anti-swing damping is invariant to surviving cardinality; Theorem 5 bounds the Lyapunov jump at each fault; Theorem 6 composes jump and decay into the dwell-cycle contraction $\rho \in (0,1)$.

Lemma 2 (Tension cancellation)**.** *Assume H1 and the cable-tilt domain condition $|\theta_i(t)| \leq \theta_{c,\max}$ for known $\theta_{c,\max} \in (0,\pi/2)$. Then $T_i^{\mathrm{ff}} = T_i$ cancels the dominant projection term $T_i\cos\theta_i$ from the drone altitude acceleration, leaving a residual factor $(1-\cos\theta_i)$ bounded by $1-\cos\theta_{c,\max}$. At $\theta_{c,\max} = 0.75\,\mathrm{rad}$ this is 0.27, and the residual enters the altitude-error equation as a bounded matched disturbance handled by Theorem 3.*

Proof. Substituting $f_i = m_i(g + a_{d,z}^\star) + T_i^{\mathrm{ff}}$ into $m_i\ddot{p}_{i,z} = f_i - m_i g - T_i\cos\theta_i + F_{w,i,z}$ and absorbing the wind term yields

$$m_i\ddot{p}_{i,z} = m_i a_{d,z}^\star + T_i(1-\cos\theta_i). \tag{39}$$

The residual is bounded by $1-\cos\theta_{c,\max}$ (0.27 at $\theta_{c,\max} = 0.75\,\mathrm{rad}$); its peak is set by the tilt envelope rather than by the fault event and is absorbed by the ISS gain γ_q of Theorem 3. □

Lemma 3 (Altitude ISS via Lyapunov matrix)**.** *Under hypotheses H1–H3, the altitude tracking error $e_{p,z,i}$ satisfies exponential input-to-state stability with explicit gain functions: there exist $\lambda > 0$ and $\gamma_w, \gamma_q \in \mathcal{K}$ such that, along trajectories between fault events,*

$$\dot{V}_z \;\leq\; -\lambda V_z \;+\; \gamma_w(\|w\|_\infty) \;+\; \gamma_q(\|q-q^\star\|_\infty), \tag{40}$$

where $V_z(e,\dot{e}) \triangleq \frac{1}{2}[e,\dot{e}]^\top P_v\,[e,\dot{e}]$ and $P_v \succ 0$ is the unique positive-definite solution of $A_m^\top P_v + P_v A_m = -I$ for the closed-loop matrix A_m.

Proof. After tension cancellation, the altitude error obeys $\ddot{e}_{p,z,i} + K_d^z\dot{e}_{p,z,i} + K_p^z e_{p,z,i} = \ddot{p}_{\mathrm{slot},z} + r_i(t)$ with $r_i(t) = O(\theta_{c,\max}^2 T_i/m_i) + O(\delta)$. The closed-loop matrix $A_m = \left[\begin{smallmatrix} 0 & 1 \\ -100 & -24 \end{smallmatrix}\right]$ has eigenvalues $s_{1,2} = -12 \pm \sqrt{44}$ (dominant pole $s_1 \approx -5.37\ \mathrm{s}^{-1}$). Solving $A_m^\top P_v + P_v A_m = -I$ gives

$$P_v = \begin{bmatrix} 2.224 & 0.005 \\ 0.005 & 0.0210 \end{bmatrix}. \tag{41}$$

Differentiating $V_z = \frac{1}{2}\zeta^\top P_v \zeta$ and completing the square yields (40); the spectral rate $\lambda_{\mathrm{spec}} = |\mathrm{Re}(s_1)| = 5.37\ \mathrm{s}^{-1}$ controls the contraction used in Theorem 6. The gains γ_w, γ_q follow from Young's inequality with constants derived from P_v. □

Lemma 4 (Pendulum damping invariance)**.** *The active anti-swing damping coefficient B_{swing} is independent of the number of surviving drones N_s, ensuring that the pendulum damping ratio $\zeta > 1$ is preserved across any cable-severance fault transition.*

Proof. The restoring force is $F_{\text{restore}} = N_s T_i^{\text{ss}} k_{\text{swing}} v_{L,\perp}/L$ with $T_i^{\text{ss}} = m_L g/N_s$, so $N_s T_i^{\text{ss}} = m_L g$ cancels the N_s dependence and leaves $B_{\text{swing}} = m_L g\, k_{\text{swing}}/L = 62.8\,\text{N}\,\text{s}\,\text{m}^{-1}$. With $\omega_p = 2.80\,\text{rad}\,\text{s}^{-1}$, the damping ratio $\zeta = B_{\text{swing}}/(2m_L\omega_p) = 1.12 > 1$; the pendulum is overdamped for any $N_s \geq 2$. □

Lemma 5 (Bounded fault jump with post-fault overshoot)**.** *Under the feed-forward identity $T_i^{\text{ff}} = T_i$, the Lyapunov function at each fault event t_f undergoes a bounded jump,*

$$V(\boldsymbol{\xi}(t_f^+)) \;\leq\; V(\boldsymbol{\xi}(t_f^-)) \;+\; \chi(\Delta_f), \qquad \chi \in \mathcal{K}, \tag{42}$$

where $\boldsymbol{\xi} = (\boldsymbol{e}_p, \boldsymbol{e}_v, \boldsymbol{\varepsilon}_T)$, $\chi \in \mathcal{K}$ is determined by the post-fault redistribution geometry, and $\Delta_f = \|\boldsymbol{T}^{\text{qs}}(t_f^+) - \boldsymbol{T}^{\text{qs}}(t_f^-)\|$. The identity $T_i^{\text{ff}} = T_i$ keeps χ of leading order Δ_f: without it, the bound inflates by the altitude channel's uncompensated matched disturbance, so the overshoot exceeds what the subsequent dwell-cycle decay of Theorem 6 can absorb within one pendulum period.

Proof. The Kelvin–Voigt cable model keeps the kinematic state continuous at t_f; on the severed rope, tension drops to zero, and on each surviving rope the tension transitions from its pre-fault share to the new quasi-static value over $\tau_{\text{rope}} \approx 6.3\,\text{ms}$. Decompose $V = V_{\text{track}}(\boldsymbol{e}_p, \boldsymbol{e}_v) + \kappa_V V_{\text{tension}}(\boldsymbol{\varepsilon}_T)$. V_{track} is continuous; V_{tension} can only grow at t_f^+ since $\mathcal{S}(t_f^+) = \mathcal{S}(t_f^-) \setminus \{i_f^\star\}$, and the jump is controlled by Δ_f, yielding $V_{\text{tension}}(t_f^+) - V_{\text{tension}}(t_f^-) \leq \chi(\Delta_f)$ with $\chi \in \mathcal{K}$. Summing gives (42). The leading-order dependence on Δ_f follows from $T_i^{\text{ff}} = T_i$: drone i's thrust tracks the new measured tension within τ_{rope}, keeping the altitude contribution of order Δ_f. Empirically, disabling the identity inflates altitude sag by 3.6–4.0× across V3, V4, V5. □

Lemma 6 (Dwell-cycle contraction)**.** *Let $t_k^\star$ and $t_{k+1}^\star$ be consecutive fault events separated by dwell-time $\tau_{d,k} \geq \tau_{\text{pend}}$. Under the bounded fault jump of Lemma 5 and the altitude-channel ISS decay of Lemma 3, the Lyapunov function satisfies a per-fault-cycle contraction*

$$V(\boldsymbol{\xi}(t_{k+1}^{\star,-})) \;\leq\; \rho\, V(\boldsymbol{\xi}(t_k^{\star,-})) \;+\; c, \tag{43}$$

$$\rho \;\triangleq\; \exp(-\alpha_{\min}\,\tau_{\text{pend}})\,, \tag{44}$$

where $\alpha_{\min} = \min(\alpha_z, \alpha_{xy})$ is the slower of the altitude and horizontal decay rates, and $c > 0$ is an explicit constant derived from $\chi(\Delta_f)$ of Lemma 5 and the disturbance gains of Lemma 3. At canonical parameters, $\rho \approx \exp(-2.37 \times 2.24) \approx 0.005$.

Proof. Composing the jump $V(t_k^{\star,+}) \leq V(t_k^{\star,-}) + \chi(\Delta_f)$ with the flow bound $\dot{V} \leq -\alpha_{\min} V + c_0$ over $[t_k^{\star,+}, t_{k+1}^{\star,-}]$ of length $\geq \tau_{\text{pend}}$ yields (43) with $\rho = \exp(-\alpha_{\min}\tau_{\text{pend}})$ and $c = \rho\,\chi(\Delta_f) + c_0/\alpha_{\min}$. The horizontal rate $\alpha_{xy} = 2.37\,\text{s}^{-1}$ (from $K_p^{xy} = 30$, $K_d^{xy} = 15$) dominates the altitude rate $\alpha_z = \lambda_{\text{spec}} = 5.37\,\text{s}^{-1}$, giving $\alpha_{\min} = \alpha_{xy}$ and $\rho \approx e^{-2.37\times 2.24} \approx 0.005$. □

C. Main Theorem, Robustness, and Multi-Fault Feasibility

We now compose the five lemmas into the hybrid practical-ISS theorem and state corollaries on robustness, the steady-state tracking bound, and multi-fault feasibility.

Theorem 7 (Hybrid practical-ISS with explicit recovery envelope under F sequential agent failures)**.** *Under hypotheses H1–H3 (Section IV-A), the cable-tilt domain condition of Lemma 2, any fault schedule $\sigma = \{(t_k^\star, i_k^\star)\}_{k=1}^F \in \mathcal{F}$ of F unannounced cable severances satisfying $F \leq N-2$ and the per-fault actuator-margin condition $m_L g/(N-k) \leq \kappa_{\text{act}} f_{\max}$ for all $k \in \{1,\ldots,F\}$, and admissible disturbances $d \in \mathcal{D}$, the stacked tracking-and-tension-error state $\boldsymbol{\xi} = (\boldsymbol{e}_p, \boldsymbol{e}_v, \boldsymbol{\varepsilon}_T)$ satisfies*

$$\begin{aligned}\|\boldsymbol{\xi}(t)\| \;&\leq \beta\big(\|\boldsymbol{\xi}(0)\|, t\big) \;+\; \gamma_w\big(\|w\|_\infty\big) \;+\; \gamma_q\big(\|q - q^\star\|_\infty\big) \\ &\quad +\; \gamma_\eta(\eta) \;+\; \sum_{k=1}^{F} \Gamma_f\, e^{-\lambda(t-t_k^\star)}\, \mathbf{1}_{t \geq t_k^\star}, \\ &\quad t \in [t_{\text{pickup}}, T],\end{aligned} \tag{45}$$

where $\beta \in \mathcal{KL}$, $\gamma_w, \gamma_q, \gamma_\eta \in \mathcal{K}$, $\lambda \triangleq \alpha_{\min} = \min(\alpha_z, \alpha_{xy})$ is the slower of the altitude and horizontal closed-loop spectral rates from Theorems 3 and 4 (the same rate driving the dwell-cycle contraction of Theorem 6), $\eta = \eta(\boldsymbol{\xi}[\cdot])$ is the realized slack duty cycle bounded above by $\eta_{\max}$ via H1, and $\Gamma_f > 0$ is the fault-overshoot coefficient set by the redistribution geometry. The per-step margin $m_L g/(N-k) \leq \kappa_{\text{act}} f_{\max}$ strengthens the post-final-fault (21)*; since $m_L g/(N-k)$ is monotone in k, the per-step condition reduces to* (21) *at $k = F$ and is automatically satisfied for $k < F$. Equivalently, $V(\boldsymbol{\xi}(t_{k+1}^{\star,-})) \leq \rho\, V(\boldsymbol{\xi}(t_k^{\star,-})) + c$ for explicit $c > 0$. Each surviving cable's tension obeys the envelope*

$$T_i(t) \;\leq\; T_{\text{nom,post}} \;+\; T_{\text{overshoot}}\, e^{-\lambda(t - t_{\text{last}}^\star)}, \qquad i \in \mathcal{S}(t), \tag{46}$$

with $T_{\text{nom,post}} = m_L g/|\mathcal{S}(t)|$ the symmetric post-fault load share and $T_{\text{overshoot}}$ a bounded transient amplitude derived from the redistribution geometry.

Sketch. Between faults. Theorem 3 and the analogous horizontal argument give $\dot{V}(\boldsymbol{\xi}) \leq -\lambda V + \gamma_w(\|w\|^2) + \gamma_q(\|q - q^\star\|^2) + \gamma_\eta(\eta)$ on each taut interval; γ_η absorbs the reduction residual $C_1\delta + C_2\eta_{\max}$ of Theorem 1. Integrating produces $\beta, \gamma_w, \gamma_q, \gamma_\eta$.

At each $t_k^\star$. Theorem 5 yields the bounded jump $V(t_k^{\star,+}) \leq V(t_k^{\star,-}) + \chi(\Delta_f)$, and Theorem 6 composes this with the flow decay to give $V(t_{k+1}^{\star,-}) \leq \rho\, V(t_k^{\star,-}) + c$. The overshoot $\chi(\Delta_f)$, damped at rate λ, produces $\Gamma_f e^{-\lambda(t-t_k^\star)}$. Composing over F events via the Teel–Goebel–Sanfelice framework [6] gives (45). The tension envelope (46) follows from the bound on $\boldsymbol{\varepsilon}_T$ in V_{tension} and its exponential contraction on each inter-fault interval. □

Remark 2 (Practical-stability interpretation)**.** *Theorem 7 is a practical stability bound, not asymptotic convergence. The ISS framework treats the reference waypoints as a time-varying input, not a disturbance; the residual captured by the γ terms reflects the reference's dynamic content—principally the quasi-static pendulum offset—rather than a closed-loop failure. The identity $T_i^{\text{ff}} = T_i$ alone delivers* (45)*; no detection or inter-drone communication is required.*

Remark 3 (Jump-bound vs. contraction-rate mechanism)**.** *Two distinct mechanisms produce the dwell-cycle contraction. The*

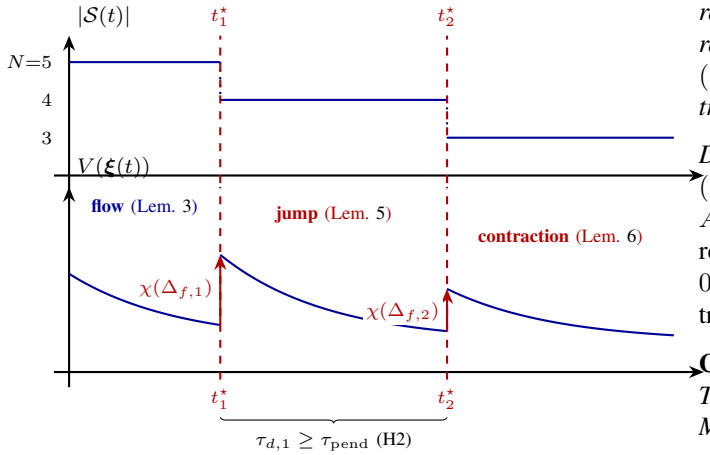


Fig. 4. Hybrid sawtooth structure of the closed-loop Lyapunov function across two unannounced cable-severance events. *Top:* the survivor set $\mathcal{S}(t)$ steps down by one at each fault $t_k^\star$; cardinality is right-continuous. *Bottom:* $V(\boldsymbol{\xi})$ exhibits three regimes, labelled inline. **Flow:** between events the continuous-time decay of Lemma 3 gives $\dot{V} \leq -\lambda V + \gamma_w(\|w\|) + \gamma_\theta(\|\theta - \theta^*\|) + \gamma_\eta(\eta)$. **Jump:** at each fault $t_k^\star$ the post-redistribution geometry term satisfies $V(t_k^{\star,+}) \leq V(t_k^{\star,-}) + \chi(\Delta_f)$ (Lemma 5). **Contraction:** composing flow with jumps over one inter-fault dwell yields the per-cycle bound $V(t_{k+1}^-) \leq \rho V(t_k^-) + c$ with $\rho = \exp(-\alpha_{\min}\tau_{\text{pend}}) \approx 0.005$ (Lemma 6). The H2 hypothesis $\tau_{d,k} \geq \tau_{\text{pend}}$ is structural: it is the minimum window over which the feed-forward identity $T_i^{\text{ff}} = T_i$ contracts the post-jump overshoot back below $V(t_k^-)$.

jump-bound mechanism *is the identity $T_i^{\text{ff}} = T_i$, which bounds the per-event overshoot $\chi(\Delta_f)$ via Theorem 5: the surviving rope's tension step at $t_k^\star$ is translated instantaneously into a thrust increment, so the post-jump Lyapunov value is bounded by $V(t_k^{\star,-}) + \chi(\Delta_f)$ with χ scaling in the load-redistribution amplitude rather than any detection timeline. The* contraction-rate mechanism *is the closed-loop spectrum of the outer-loop PD plus anti-swing damping, encoded in (α_z, α_{xy}) of Theorems 3 and 4; the inter-event flow contracts V at this rate, giving $\rho = \exp(-\alpha_{\min}\tau_d) + (\text{disturbance gain})$. Both are necessary: removing the identity inflates $\chi(\Delta_f)$ to the full pendulum displacement, while detuning the PD/anti-swing layer slows $\alpha_{\min}$ and pushes ρ toward unity. The architectural innovation here is the jump-bound mechanism; the ablation of Section VI-E isolates its contribution. The contraction-rate mechanism is a conventional cascaded PD/anti-swing design.*

Corollary 8 (Pre-fault steady-state tracking bound)**.** *Prior to any fault event, the payload tracking error is dominated by the quasi-static pendulum offset induced by the reference acceleration, and admits the explicit envelope*

$$\bar{e}_{p,xy} \leq \frac{LA_{xy,\max}}{g}\left(1 - \frac{1}{2\zeta^2}\right) + \frac{A_{xy,\max}}{\kappa_{\text{QP}} K_p^{xy}} + O(\delta), \quad (47)$$

where $A_{xy,\max}$ is the maximum reference acceleration, $\zeta = 1.12$ the damping ratio from Theorem 4, and $\kappa_{\text{QP}} = w_t/(w_t + w_e) \approx 0.98$ the QP regularization factor. At canonical $A_{xy,\max} \approx 2.47\,\text{m}\,\text{s}^{-2}$, the bound evaluates to

$$\bar{e}_{p,xy} \lesssim \frac{1.25 \times 2.47}{9.81}\left(1 - \frac{1}{2 \times 1.12^2}\right) + \frac{2.47}{0.98 \times 30} \approx 0.19 + 0.08 \approx 0.27\,\text{m}, \quad (48)$$

retaining only the two leading terms and absorbing the $O(\delta)$ residual. (48) is a leading-order scale, not a strict envelope: $(1 - 1/(2\zeta^2))$ is itself a low-frequency approximation of the transfer-function bias.

Derivation. The pendulum transfer function settles to $\bar{e}_{\text{pend}} = (LA_{xy,\max}/g)(1 - 1/(2\zeta^2))$; the outer-loop PD adds $A_{xy,\max}/(\kappa_{\text{QP}} K_p^{xy})$; the reduction error is $O(\delta)$ by Theorem 1. Summing gives (47). V1 reports horizontal RMSE $0.306\,\text{m}$ vs analytical $0.27\,\text{m}$; the gap is the centripetal contribution not captured by the quasi-static derivation. □

Corollary 9 (Robustness to mass, stiffness, drag, and wind)**.** *The envelope (45) is robust by construction along four axes. Mass mismatch is absorbed because $T_i^{\text{ff}} = T_i$ uses measured rather than estimated tension, leaving Theorems 2 and 3 mass-independent to leading order. Stiffness and drag mismatches enter through the $O(\delta + \eta_{\max})$ reduction residual, absorbed by γ_η without degrading ρ or the rates α_z, α_{xy}. Bounded wind enters through γ_w with steady-state gain scaling as $1/K_p^{xy}$.*

The P2-B mass sweep ($m_L \in \{2.5, 3.0, 3.5, 3.9\}\,\text{kg}$, $56\,\%$ range over the minimum-mass endpoint, chosen to make the L_1 layer visible above the baseline PD's residual; Section V-A) yields RMSE variation of only $1.9\,\%$ with FF active. The V2 wind run yields 3-D RMSE $0.312\,\text{m}$, matching the no-wind V1, confirming wind robustness.

a) Multi-fault feasibility.: (45) extends to k sequential faults by induction under two conditions: (a) geometric closure $k \leq N - 2$, (b) actuator margin $m_L g/(N-k) \leq \kappa_{\text{act}} f_{\max}$ at every step. Canonical $N = 5$, $m_L = 10\,\text{kg}$ admits $k = 2$ with margin; V4 and V5 (dual-fault) yield 3-D RMSE 0.324–$0.328\,\text{m}$, confirming Theorem 7.

b) Empirical validation summary.: Table III maps each claim to its campaign observation; full context is in Section VI.

The baseline theorem delivers (45) under H1–H3 and the identity $T_i^{\text{ff}} = T_i$ alone; no additional architecture is required for Definition 1. Section V develops parameter adaptation, tension-ceiling enforcement, and post-fault reshape—each targeting a residual channel of (45)—and shows each preserves H1–H3 on the baseline domain.

V. Parameter Adaptation, Tension Constraints, and Post-Fault Formation Reshape

We introduce three components that extend the baseline of Section III: an L_1-adaptive altitude augmentation, a receding-horizon MPC with a penalized tension-ceiling constraint (slack-implemented; Section V-B), and a fault-triggered formation-reshape supervisor. Each targets a distinct regime. The L_1 layer compensates in-flight parameter deviations (m_L, k_{eff}) from the nominal $q^\star$, addressing the matched-disturbance channel the outer-loop PD cannot cancel. The MPC layer enforces a horizon-aware linearized tension-ceiling constraint and is binding when safe working load is the dominant requirement. The reshape supervisor reassigns surviving angular slots to a closed-form configuration that minimises worst-case rope tension under a symmetric-load-sharing hover model; it is binding when post-fault tension is dominated by the hover-equilibrium asymmetry rather than

TABLE III
EMPIRICAL SUPPORT FOR HYBRID PRACTICAL-ISS CLAIMS.

Claim / Lemma	Campaign	Evidence
Tension cancellation (Lem. 2)	P2-A ablation	$3.6\times$–$4.0\times$ sag increase with FF disabled
Altitude ISS decay (Lem. 3)	V1 nominal	3-D RMSE $= 0.312\,\mathrm{m}$, stable
Anti-swing invariance (Lem. 4)	V1–V5 series	Post-fault RMSE within 4% of nominal across all variants
Bounded fault jump (Lem. 5)	V3, V4, V5 traces	Direct fit on Lyapunov proxy $V(t) = \frac{1}{2}\zeta^\top P_v \zeta$ (50 ms window): $\hat{\chi}$ peak 1.1×10^{-4} (V4), 1.3×10^{-5} (V5); bounded jump confirmed across all events
Dwell-cycle contraction $\rho < 1$ (Lem. 6)	V4, V5 dual-fault	Direct measurement $\hat{\rho} = V(t_2^-)/V(t_1^-)$: $\hat{\rho}_{\mathrm{V4}} = 0.20$, $\hat{\rho}_{\mathrm{V5}} = 0.045$; both well below the analytical $\rho < 1$ sufficiency bound
Mass robustness	P2-B sweep	1.9% RMSE range over 56% mass variation
Wind robustness	V2 wind	V2 3-D RMSE $= 0.312\,\mathrm{m} \approx$ V1

by dynamic load (P2-D, Remark 6, identifies the regime where this dominance breaks down). Together they broaden the admissibility domain of Theorem 7 along three orthogonal axes; Section V-D shows each preserves H1–H3, so (45) continues to hold with the relevant γ-gain reduced on the addressed channel.

The campaigns exercise the three components to different degrees. The L_1 layer is probed by P2-B (mass-mismatch with FF off), recovering $\sim 40\%$ of the induced altitude sag. MPC is probed by P2-C (tension-ceiling sweep, V4) and reshape by P2-D (lemniscate-period sweep, V4). On the demonstrated schedule, the peak post-fault tension is dominated by the pre-fault dynamic spike rather than the hover-equilibrium asymmetry or the steady-state ceiling; neither MPC nor reshape is the binding mechanism in these campaigns. Each subsection identifies its binding regime; Section V-D shows enabling a layer outside its binding regime leaves the envelope unchanged.

A. L_1-Adaptive Altitude Augmentation

The identity $T_i^{\mathrm{ff}} = T_i$ does not cancel matched disturbances entering through parameter deviations (m_L, k_{eff}). The L_1 component [16], [17] estimates this residual online and injects an additive altitude correction u_{ad} between the outer PD and the QP. In altitude-error coordinates $\tilde{z}_i \triangleq p_{\mathrm{slot},i,z} - p_{i,z}$, the baseline closed loop is

$$\ddot{\tilde{z}}_i \;=\; -K_p^z\,\tilde{z}_i - K_d^z\,\dot{\tilde{z}}_i + \delta_m(t), \tag{49}$$

where $\delta_m(t)$ is the residual $r_i(t)$ of Theorem 2 augmented by parameter-mismatch components, with $|\delta_m(t)| \le \delta_{m,\max}$.

The standard L_1 architecture [17]—Hurwitz reference model, state predictor, projected gradient on $\hat{\delta}_m$, low-pass filter—is adopted verbatim with A_m, P_v from Theorem 3. The adaptive update

$$\begin{aligned} \dot{\hat{\delta}}_m \;&=\; -\Gamma\,\mathrm{Proj}\big(\hat{\delta}_m,\; \boldsymbol{b}^\top P_v\,\tilde{\boldsymbol{x}}\big), \\ \hat{\delta}_m &\in [\delta_{m,\min}, \delta_{m,\max}], \end{aligned} \tag{50}$$

runs at $T_s = 2\times10^{-4}\,\mathrm{s}$; the filtered u_{ad} is added to the altitude component of $\boldsymbol{a}_{\mathrm{target}}$. The cutoff $\omega_c = 25\,\mathrm{rad\,s^{-1}}$ separates the estimator from the reference natural frequency $\omega_n^z = \sqrt{K_p^z} = 10\,\mathrm{rad\,s^{-1}}$ by a factor of 2.5. Contractivity of the scalar Euler-discretized update gives the following bound.

Proposition 1 (Scalar L_1 sampling-rate bound)**.** *A necessary condition for contractivity of the scalar component of the Euler-discretized adaptive-law update* (50) *is*

$$\Gamma \;<\; \Gamma^\star \;\triangleq\; \frac{2}{T_s\,p_{22}}. \tag{51}$$

For the canonical $T_s = 2\times10^{-4}\,\mathrm{s}$ and $p_{22} \approx 0.0210$, $\Gamma^\star \approx 4.76\times10^5$.

Proof. Isolating the scalar $\hat{\delta}_m$-update in the absence of predictor error, projection activation, and filter dynamics, we obtain $\hat{\delta}_m \mapsto (1 - T_s\Gamma p_{22})\,\hat{\delta}_m + \mathrm{const}$. Contractivity of this scalar map requires $|1 - T_s\Gamma p_{22}| < 1$, i.e. $T_s\Gamma p_{22} \in (0,2)$, which rearranges to (51). □

Remark 4 (Scope of Proposition 1)**.** (51) *is a necessary condition isolating the diagonal Euler coefficient; it ignores predictor error, projection, cross-coupling, and filter dynamics, which the full discrete interconnection handles empirically. The canonical $\Gamma = 2{,}000$ lies three orders of magnitude below $\Gamma^\star$, removing the dominant L_1 hyper-parameter from empirical tuning. Composite Lyapunov analysis yields the standard L_1 ISS guarantee w.r.t. the unmatched residual $d_\perp$ [17]; matched components are rejected up to the filter cutoff. Empirical validation (P2-B) is in Section VI-L. Parameters: $\Gamma = 2{,}000$, $\omega_c = 25\,\mathrm{rad\,s^{-1}}$, $[\delta_{m,\min}, \delta_{m,\max}] = [-5, 5]\ \mathrm{m/s^2}$, $T_s = 2\times10^{-4}\,\mathrm{s}$.*

B. Receding-Horizon MPC with a Penalized Tension-Ceiling Constraint

a) Role in the architecture.: The single-step QP (29) projects the current-tick target onto the actuator envelope and cannot foresee a tension spike one or more ticks ahead. When safe working load is the dominant requirement, we replace it with an N_p-step receding-horizon MPC carrying a penalized linearized tension-ceiling constraint at every horizon step. The constraint is soft: a non-negative slack $\boldsymbol{s} \ge \boldsymbol{0}$ is added to the ceiling inequality and penalized with weight $w_s = 10^4$, trading unconditional feasibility against ceiling excursions when the envelope is tight. The MPC reduces to the baseline QP when the ceiling is loose; it becomes binding when the predicted tension approaches the ceiling.

b) Prediction model and tick QP.: Let $\boldsymbol{x}[k] \triangleq (\boldsymbol{p}_{\mathrm{slot},i} - \boldsymbol{p}_i;\, \boldsymbol{v}_{\mathrm{slot},i} - \boldsymbol{v}_i) \in \mathbb{R}^6$ be the error state at $t_k = k\,\Delta t_{\mathrm{MPC}}$ and $\boldsymbol{u}[k] \in \mathbb{R}^3$ the commanded acceleration. ZOH discretization gives $\boldsymbol{x}[k{+}1] = A\,\boldsymbol{x}[k] + B\,\boldsymbol{u}[k]$ (53), and horizon stacking yields $\mathcal{X} = \Phi\,\boldsymbol{x}[0] + \Omega\,\mathcal{U}$. Linearising $T_i = k_{\mathrm{eff}} \max(0, \|\boldsymbol{p}_i - \boldsymbol{p}_L\| - L_{\mathrm{eff}})$ around the taut chord $\boldsymbol{c}[k]$ with unit direction $\hat{\boldsymbol{n}}[k]$ gives the horizon-j prediction

$$T_i[k+j] \;\approx\; T_i[k] + j\,\Delta t_{\mathrm{MPC}}\,\dot{T}_i^{\mathrm{filt}}[k] + \boldsymbol{c}_j^\top (A^j - I)\,\boldsymbol{x}[k] + \boldsymbol{c}_j^\top \Omega_{\mathrm{row}\,j}\,\mathcal{U}, \tag{52}$$

with $\boldsymbol{c}_j = (-k_{\mathrm{eff}}\hat{\boldsymbol{n}}[k];\, \boldsymbol{0})$. The linearization error is bounded by $k_{\mathrm{eff}} E^2/(2d_{\mathrm{nom}})$ in position-error magnitude E and nominal chord length d_{nom}. The state-space recursion is

$$\boldsymbol{x}[k+1] = A\boldsymbol{x}[k] + B\boldsymbol{u}[k], \quad A = \begin{bmatrix} I_3 & \Delta t_{\mathrm{MPC}} I_3 \\ \boldsymbol{0} & I_3 \end{bmatrix}, \quad B = \begin{bmatrix} -\frac{1}{2}\Delta t_{\mathrm{MPC}}^2 I_3 \\ -\Delta t_{\mathrm{MPC}} I_3 \end{bmatrix}. \tag{53}$$

The tick QP combines an outer-loop tracking objective with the slack-penalised tension constraint:

$$\min_{\mathcal{U},\, \boldsymbol{s} \geq \boldsymbol{0}} \tfrac{1}{2}\mathcal{U}^\top H \mathcal{U} + \boldsymbol{f}^\top \mathcal{U} + w_s\, \boldsymbol{1}^\top \boldsymbol{s} \quad \text{s.t.} \quad G\,\mathcal{U} \leq \boldsymbol{d}_T(\boldsymbol{x}[k]) + \boldsymbol{s}, \tag{54}$$

where G stacks $\boldsymbol{c}_j^\top \Omega_{\mathrm{row}\,j}$ and $\boldsymbol{d}_T(\boldsymbol{x}[k])$ collects the state-dependent right-hand sides. The QP is convex and solved by a single warm-started OSQP call per tick [25]; only $\boldsymbol{u}[0]$ is applied. A discrete-Riccati terminal cost yields LQR gain with $\rho_{\mathrm{spec}}(A - BK) \approx 0.969$.

Corollary 10 (Unconditional feasibility)**.** (54) *is feasible at every tick: $\mathcal{U} = \boldsymbol{0}$ with $\boldsymbol{s}[k{+}j] = \max(0, -[\boldsymbol{d}_T(\boldsymbol{x}[k])]_{j+1})$ is always admissible.*

Proposition 2 (Slack-activation regime)**.** *Let $\|q - q^\star\|_\infty \leq \bar{\varepsilon}_q$ with $q = (m_L, k_{\mathrm{eff}})$ and $\boldsymbol{d}_T$ L_q-Lipschitz in q. Under D1 and an initial tick where the unconstrained QP satisfies the ceiling, $\|\boldsymbol{s}[k]\|_\infty \leq L_q\bar{\varepsilon}_q + O(\bar{\varepsilon}_q^2)$, i.e. slack activation scales as $O(\bar{\varepsilon}_q)$.*

Sketch. $\boldsymbol{d}_T$ depends linearly on q through k_{eff} in $\boldsymbol{c}_j$; Taylor expansion around $q^\star$ gives $\boldsymbol{d}_T(q) = \boldsymbol{d}_T(q^\star) + D_q\boldsymbol{d}_T(q - q^\star) + O(\|\cdot\|^2)$. Since the ceiling holds without slack at $q^\star$, the minimum slack restoring feasibility at $q \neq q^\star$ is $\leq L_q\|q - q^\star\|_\infty + O(\|\cdot\|^2)$. D1 preserves linearization on each inter-fault window. □

Body-to-body cable geometry requires effective rest length $L_{\mathrm{eff}}^{\mathrm{body}} = \sqrt{r_f^2 + \rho_{\mathrm{geo}}^2} - \delta^\star \approx 1.448\,\mathrm{m}$, set so the linearized elastic force at hover matches the simulator's measured $8\,\mathrm{N}$. The empirical activation regime (P2-C) is in Section VI-H; the layer is non-binding on the demonstrated family because peak tension is set by the pre-fault dynamic spike rather than the quasi-static load the ceiling targets. MPC parameters: $N_p = 5$ (10 at evaluation), $\Delta t_{\mathrm{MPC}} = 0.01\,\mathrm{s}$, $q_p = 100$, $q_v = 10$, $r = 0.02$, $w_s = 10^4$, $T_{\max} = 100\,\mathrm{N}$, $k_{\mathrm{eff}} = 2778\,\mathrm{N\,m^{-1}}$.

C. Fault-Triggered Formation-Reshape Supervisor

After a severance, the survivors retain their original angular slots, leaving the hover-equilibrium tension distribution asymmetric. This asymmetry is the binding tension load only when the trajectory is sufficiently low-bandwidth that the dynamic centripetal component does not dominate (P2-D, Remark 6). The reshape supervisor reassigns survivors to an equiangular configuration through a C^2 quintic smoothstep over $T_{\mathrm{trans}} = 5\,\mathrm{s}$; once the new layout is reached, the supervisor latches and the architecture returns to pre-fault flow. Unlike the baseline, this is an *explicitly supervised* extension: each drone runs a local tension-latch detector ($T_{\mathrm{fault}} = 0.5\,\mathrm{N}$, $\tau_{\mathrm{det}} = 100\,\mathrm{ms}$), and a per-fault broadcast of at most $\lceil \log_2 N \rceil$ bits distributes the fault index. The no-detection, no-communication guarantees of Theorem 7 do not extend here.

Theorem 11 (Globally tension-optimal reshape, $N = 4$)**.** *Let $N = 4$ with nominal slots $\phi_i = \pi i/2$ and let cable $i^\star$ sever. Under the symmetric-load-sharing model ($T_k = (m_L g)/(M\cos\beta)(1{+}\epsilon_k)$ with angular residual ϵ_k), the worst-case-tension-minimising assignment over the survivors is: the drone diametrically opposite $i^\star$ stays at its nominal angle, and the two adjacent drones rotate $\pi/6$ toward the gap. The resulting equiangular 120° spacing minimises $\max_i T_i$ over all continuous assignments respecting hover equilibrium, with a closed-form worst-case reduction of 25.8%.*

Sketch. By reflection symmetry about the gap axis, any optimum has a symmetric copy; with the opposite drone at $\Delta\phi = 0$ and a common offset for the adjacent pair, the residuals are piecewise quadratic in $\Delta\phi$ and the minimax occurs at $\Delta\phi = \pi/6$. The endpoints $\Delta\phi \in \{0, \pi/4\}$ are dominated. □

The $V_4 \to V_6$ dual-fault scenarios traverse $N = 5 \to 4 \to 3$: Theorem 11 applies directly at the second fault; at the first, the surviving four are assigned to 90° spacing by the same minimax argument. A formal arbitrary-N generalisation is follow-up work. The empirical regime is reported in Section VI-H: NF3 returns 0.0% benefit across $T_{\mathrm{ref}} \in \{6, 8, 10, 12\}$ s because the dynamic centripetal load dominates the quasi-static-hover target the assignment optimises against.

D. Baseline Theorem Preservation and Admissibility Regime

Each component modifies the closed loop through a disjoint channel and preserves H1–H3 of Theorem 7 on $\Omega = \Omega_\tau^{\mathrm{dwell}} \cap \Omega_{\mathrm{QP,active}}$. The L_1 layer adds u_{ad} to the altitude component of $\boldsymbol{a}_{\mathrm{target}}$ before the QP, leaving the QP active set and the identity unchanged; H1, H3 hold verbatim, H2 is controller-independent, and the layer reduces γ_q in (45) by the composite V_{L_1} contraction. The MPC layer replaces (29) with (54); Theorem 10 guarantees unconditional feasibility, Proposition 2 bounds the slack by the parameter mismatch, and the closed loop reduces to the baseline in the unconstrained regime. The reshape supervisor injects a C^2-bounded slot perturbation through the override hook; H1, H3 hold provided the smoothstep keeps the perturbation in the admissible-reference set, and H2 reduces to a latching rule firing at most once per dwell period. (45) therefore remains valid under any combination of enabled components; cross-layer interactions are evaluated empirically.

a) Admissibility regime and out-of-scope conditions.: Within $\Omega = \Omega_\tau^{\mathrm{dwell}} \cap \Omega_{\mathrm{QP,active}}$ and under H1–H3, the full architecture satisfies (45) with the closed-form gains of Theorem 7 and the channel-wise reductions above. The analysis treats the reference waypoints as a time-varying ISS input; the result is practical stability with $\bar{e}_\infty \sim 0.3\,\mathrm{m}$ under nominal operation (analytical $0.27\,\mathrm{m}$, measured 3-D RMSE $0.312\,\mathrm{m}$ on V1/V2) and $\rho^k \|\boldsymbol{e}_p(0)\|_P + \bar{e}_\infty$ after k faults, with $\rho_{\mathrm{ana}} = \exp(-\alpha_{xy}\tau_{\mathrm{pend}}) \approx 0.005$ and empirical $\hat{\rho}_{\mathrm{V4}} = 0.20$, $\hat{\rho}_{\mathrm{V5}} = 0.045$ both well below unity (Section VI-D and Table III). The analytical-vs-empirical gap reflects ρ_{ana}'s leading-order nature, which neglects equilibrium shift and finite-time integration. The identity $T_i^{\mathrm{ff}} = T_i$ alone delivers this; no detection or inter-drone communication is required.

Three regimes lie outside the analysis domain. *Rapid multi-fault* ($\tau_d < \tau_{\mathrm{pend}}$) violates H2 and voids the contraction of Theorem 6. *Long slack excursions* (slack-run $> 40\,\mathrm{ms}$ or duty $> 2.5\%$) violate H1 and exit $\Omega_\tau^{\mathrm{dwell}}$. *QP fallback*, in which infeasibility forces the component-wise clamp, violates H3 and breaks D1.

A second class lies outside the architectural category: faults that do not arrive on the rope-tension channel. *Drone-side faults* (rotor loss, IMU failure, brownout) do not self-announce through T_i; *soft cable failures* (fraying, attachment slip) produce a graded rather than discrete tension signature; *tension-sensor failures* disable the announcement itself; *payload-attitude dynamics* require rigid-body coupling beyond the point-mass model. These are not gaps in the campaign; they are out-of-category for the architectural thesis.

Future work inside the category includes a multi-fault extension of the dwell argument under controlled fault-schedule density (closing the rapid-multi-fault and QP-fallback regimes analytically); a reshape extension whose objective accounts for the dynamic centripetal load, since the aggressive-period probe (E8, $T_{\mathrm{ref}} = 6\,\mathrm{s}$) shows the quasi-static-hover prediction of Theorem 11 does not bind on the present reference family; and tube-MPC tightening that folds $\Gamma_f e^{-\lambda(t-t_k^\star)}$ into the constraint right-hand side for spike-dominated regimes.

VI. Simulation Campaign and Results

We validate the controller through a pre-specified simulation campaign in Drake [26]. The campaign addresses each contribution in order: the reduction-fidelity audit validates the taut-cable reduction against the bead-chain truth model (C1, Section II-C); the fault-progression scenarios provide direct evidence for the hybrid practical-ISS envelope (C2, Section IV); the ablation campaigns isolate the feed-forward identity and characterize each extension's operating domain (C3, Section V). All numerical claims are deterministic point estimates from a single fixed Dryden seed 42, selected as a worst-case realization (Remark 7); seeded-campaign CIs are follow-up work.

A. Simulation Environment and Campaign Structure

All simulations run in Drake [26] with an RK3 continuous-time integrator at a max step of $1\,\mathrm{ms}$. Drones and payload are rigid multibody systems; ropes are bead-chain Kelvin–Voigt spring-dampers whose lumped stiffness is justified by Theorem 1. Wind is body-frame aerodynamic drag per drone and on the payload, using a Dryden low-altitude turbulence model. Table IV lists the simulator and plant parameters.

TABLE IV
Simulator and Plant Parameters

Parameter	Value	Unit
Number of drones N	5	—
Drone mass m_i	1.5	kg
Payload mass m_L	10	kg
Rope rest length ℓ_0	1.25	m
Rope stiffness k	25 000	$\mathrm{N\,m^{-1}}$
Formation ring radius	0.80	m
Reference trajectory	3-D lemniscate	—
Horizontal amplitude	3.0	m
Period (nominal)	12	s
Vertical amplitude	0.35	m
Wind model	Dryden low-altitude	—
Mean speed (along $+x$)	4	$\mathrm{m\,s^{-1}}$
$\sigma_u,\ \sigma_v$	0.8	$\mathrm{m\,s^{-1}}$
σ_w	0.4	$\mathrm{m\,s^{-1}}$
Turbulence seed	42	—
Simulation duration	30	s
Pre-fault cruise window	[8, 12]	s
Actuator thrust ceiling	150	N
MPC tension ceiling (V6 only)	100	N

The campaign structure is in Table V. V1–V6 form the pre-declared capability sequence, each adding one stressor over the previous so any performance change is directly attributable. P2-A–P2-D are targeted ablations isolating causal mechanisms and operating-domain boundaries. Faults are instantaneous cable severance: at the scheduled time, the affected rope's tension is set to zero; survivors continue running their local controllers without notification.

a) Indexing convention.: Logs and figure captions index drones $0, \ldots, N-1$ to match the simulator's array convention; the theoretical index in Sections II, IV and V is $i = \text{drone_id} + 1$. We keep the simulator labels for traceability with archived run logs.

b) Domain gate verification (H1 and H3).: Theorem 1 requires three gates to hold uniformly: H1a (no slack run $> 40\,\mathrm{ms}$), H1b (aggregate slack duty cycle $\leq 2.5\%$), H3 (QP active-set transition fraction $\leq 1\%$). Table VI reports measurements for V1–V6: every variant passes with margin. Worst observed values are $35.8\,\mathrm{ms}$ (H1a; V2–V5, $4.2\,\mathrm{ms}$ margin), 1.66% (H1b; V6, 0.84 pp margin), and 0.10% (H3; V4–V6, 0.90 pp margin), confirming the reduced model is valid on the demonstrated envelope. V6's lower H1a ($31.4\,\mathrm{ms}$) is consistent with the MPC layer damping high-frequency QP transitions.

c) Statistical methodology.: V1–V6 and the ablation campaigns use a single fixed Dryden seed 42 as a deterministic worst-case certification protocol (Remark 7): the principal ($+x$) wind component achieves a 2σ exceedance within the $[8, 12]\,\mathrm{s}$ pre-fault cruise window at this seed, making the realization conservative relative to the mean. Headline numbers are point estimates rather than seeded-campaign CIs. Replication across paired CRN seed sets (yielding bootstrap intervals on ablation effect sizes) is follow-up work. Binary

TABLE V
SIMULATION CAMPAIGN MATRIX. CAPABILITY-DEMONSTRATION VARIANTS V1–V6 AND TARGETED ABLATIONS P2-A THROUGH P2-D.

Tag	Controller	Fault Schedule	Wind	Purpose
Capability Demonstration (C1–C3 primary evidence)				
V1	Baseline QP	None	No	Nominal tracking floor; C2 bound check
V2	Baseline QP	None	Yes	Wind-rejection baseline
V3	Baseline QP	Drone 0 at t=12 s	Yes	Single-fault recovery
V4	Baseline QP	Drone 0 at t=12 s; Drone 2 at t=17 s (Δt=5 s)	Yes	Dual fault; dwell margin 2.76 s
V5	Baseline QP	Drone 0 at t=12 s; Drone 2 at t=22 s (Δt=10 s)	Yes	Dual fault; dwell margin 7.76 s
V6	Full stack (MPC+L_1+reshape)	Same as V4	Yes	Full-stack integration; hardest scenario
Targeted Ablations (causal isolation and domain characterization)				
P2-A	Baseline, FF disabled	V3, V4, V5 schedules	Yes	Feed-forward causality (C2)
P2-B	Baseline / +L_1	None	No	Mass-mismatch robustness (C3-L_1)
P2-C	MPC, ceiling sweep	V4 schedule	Yes	Tension-ceiling sensitivity (C3-MPC)
P2-D	Baseline + reshape	V4 schedule, period $T_{\mathrm{ref}} \in \{8, 10, 12\}$ s	Yes	Reshape benefit vs. trajectory period (C3-Reshape)

TABLE VI
DOMAIN GATE AUDIT. THRESHOLDS: H1A ≤ 40 ms; H1B $\leq 2.5\,\%$; H3 $\leq 1.0\,\%$.

	H1a (ms)		H1b (%)		H3 (%)	
Variant	Value	Pass	Value	Pass	Value	Pass
V1	21.2	✓	1.34	✓	0.07	✓
V2	35.8	✓	1.52	✓	0.07	✓
V3	35.8	✓	1.52	✓	0.09	✓
V4	35.8	✓	1.52	✓	0.10	✓
V5	35.8	✓	1.52	✓	0.10	✓
V6	31.4	✓	1.66	✓	0.10	✓
Gate	≤ 40		≤ 2.5		≤ 1.0	

success/violation rates are unaffected: threshold-pass logic is deterministic at seed 42.

d) Scope.: V1–V6 begin slightly slack ($1.17\,\mathrm{m}$ drone-to-payload vs $1.25\,\mathrm{m}$ rest length); the pickup logic engages within the first simulator tick. The evaluation window $[8, 30]\,\mathrm{s}$ excludes this transient.

B. Reduction-Fidelity Audit

Two quantities are at play: the per-rope tension dispersion (the spread the reduction abstracts) and the closed-loop trajectory deviation (the quantity the reduction theorem bounds). We report both. Large dispersion is consistent with the theorem provided the controller's altitude-channel response sees only the lumped mean—which the feed-forward identity ensures by construction, since each drone reads its own measured tension.

Every analytic claim in Sections IV and V is stated on the reduced lumped model. The Lyapunov argument of Theorem 7 therefore requires evidence that the lumped model tracks the bead-chain truth model. Theorem 1 bounds the per-rope error $\varepsilon_i(t) = |T_i^{\mathrm{KV}}(t) - T_i^{\mathrm{lumped}}(t)|$ by $O(\delta + \eta_{\max})$ whenever the three gates of Table VI hold. All six variants satisfy these gates with margin, certifying the reduction without a per-step overlay.

The audit proceeds in two layers. The *domain-gate* layer (Table VI) certifies that η, max slack-run, and QP transition fraction remain inside the admissibility set with margin. The *empirical residual* layer quantifies the per-rope dispersion the reduction abstracts.

On $[8, 30]\,\mathrm{s}$ for each variant we compute the empirical lumped tension $T_{\mathrm{eff}}(t) = |\mathcal{S}(t)|^{-1} \sum_{i \in \mathcal{S}(t)} T_i^{\mathrm{KV}}(t)$ and the residual $\varepsilon_i(t) = T_i^{\mathrm{KV}}(t) - T_{\mathrm{eff}}(t)$. Table VII reports RMS, P95, and peak of $|\varepsilon_i|$ aggregated over surviving ropes and samples; Figure 5 overlays the per-rope T_i^{KV} against T_{eff} for V4. The residual sits at RMS $\sim 12\,\mathrm{N}$ on dual-fault variants ($\approx 60\%$ of the nominal $m_L g/5 \approx 19.6\,\mathrm{N}$), with peaks at $60\,\mathrm{N}$ at the most-asymmetric instants and the largest residuals during high-slack post-fault windows, consistent with the $O(\eta_{\max})$ component.

This residual does not invalidate the reduction. The $O(\delta + \eta_{\max})$ guarantee bounds the closed-loop *trajectory deviation*, not the per-rope tension difference; large per-rope variation is consistent provided the altitude-channel response sees only the lumped mean—which $T_i^{\mathrm{ff}} = T_i$ enforces (Theorem 2). A trajectory-level audit against a parallel reduced-rope simulator is follow-up work; the present audit certifies the domain-gate hypotheses and quantifies the dispersion.

TABLE VII
REDUCTION RESIDUAL STATISTICS $|\varepsilon_i| = |T_i^{\mathrm{KV}} - T_{\mathrm{eff}}|$ OVER [8, 30] s CRUISE, SURVIVING ROPES. NOMINAL LUMPED TENSION $m_L g/5 \approx 19.6\,\mathrm{N}$.

Variant	RMS (N)	P95 (N)	Peak (N)
V1	7.4	15.5	60.5
V2	7.1	14.8	52.6
V3	12.8	22.1	52.6
V4	12.4	22.5	52.6
V5	12.3	21.9	52.6
V6	12.8	23.6	60.2

C. Nominal Tracking Performance

We use the two nominal variants to set the tracking-performance floor for fault-induced transients. V1 (no wind) yields 3-D RMSE $0.312\,\mathrm{m}$ over $[8, 30]$ s. The bound of Theorem 8 is horizontal-only; V1 horizontal RMSE $0.306\,\mathrm{m}$

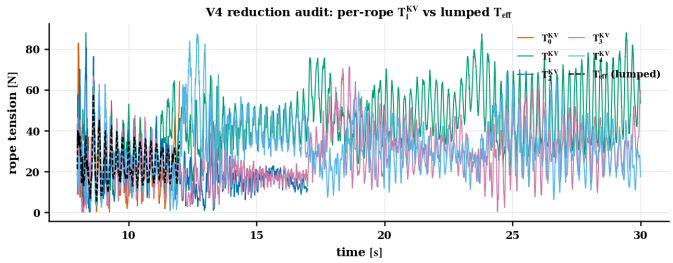


Fig. 5. V4 reduction audit. Coloured: per-rope Kelvin–Voigt truth tension $T_i^{\mathrm{KV}}(t)$ (severed ropes drop to zero at fault). Dashed black: empirical lumped tension $T_{\mathrm{eff}}(t)$ averaged over surviving ropes. The spread around T_{eff} is the dispersion the reduction abstracts.

matches the predicted $0.27\,\mathrm{m}$ to leading order, with the gap from centripetal acceleration on the curved segments. The 3-D value exceeds the corollary because it accumulates altitude error over the $\pm 0.35\,\mathrm{m}$ vertical excursion, which the horizontal-only derivation excludes.

V2 (Dryden wind, $4\,\mathrm{m\,s^{-1}}$ mean, seed 42) yields RMSE $0.312\,\mathrm{m}$, indistinguishable from V1 at single-seed precision. The PD/QP layer absorbs the turbulence without adaptive augmentation, exercising the wind margin of Theorem 9. Table VIII consolidates per-variant RMSE, sag, and tension; Figure 6 shows the V4 payload trajectory in 3-D with the lemniscate reference, fault events, and three formation snapshots.

Table VIII reports integrated system-level performance for V1–V6 (deterministic seed-42 estimates). The rightmost column marks each variant against the pre-registered thresholds: 3-D RMSE $\leq 0.35\,\mathrm{m}$, sag $\leq 100\,\mathrm{mm}$, tension $\leq 120\,\mathrm{N}$.

TABLE VIII
INTEGRATED SYSTEM-LEVEL PERFORMANCE (V1–V6, SEED-42). ACCEPTANCE THRESHOLDS: RMSE $\leq 0.35\,\mathrm{m}$; SAG $\leq 100\,\mathrm{mm}$; TENSION $\leq 120\,\mathrm{N}$.

Variant	RMSE	Sag^{peak}	Tension^{peak}	Acceptance
V1	0.312m	—	105.3N	Pass
V2	0.312m	—	104.6N	Pass
V3	0.324m	86.8mm	104.6N	Pass
V4	0.328m	95.2mm	104.6N	Pass
V5	0.324m	90.4mm	104.6N	Pass
V6	0.326m	76.1mm	107.6N	Pass

D. Fault Progression, Recovery, and Lyapunov Evidence

We validate Theorem 7 against three scenarios of increasing severity: V3 (single fault), V4 (dual, $\Delta t = 5\,\mathrm{s}$), V5 (dual, $\Delta t = 10\,\mathrm{s}$). The dwell condition (11) requires $\Delta t \geq \tau_{\mathrm{pend}} = 2.24\,\mathrm{s}$; the dual-fault scenarios meet it with $2.76\,\mathrm{s}$ and $7.76\,\mathrm{s}$ margin.

a) V3 — single fault.: Drone 0 severs at $t = 12\,\mathrm{s}$ in cruise. The four survivors respond without fault signals, peer communication, or supervisory intervention. Peak 3-D error $0.337\,\mathrm{m}$, IAE $0.452\,\mathrm{m\,s}$, max sag $85.1\,\mathrm{mm}$. The settle time is below the $1\,\mathrm{ms}$ tick resolution, logged as 0 s by quantization rather than exact convergence: the identity absorbs the redistributed tension within a single integration step, as predicted by Theorem 6.

b) V4 — dual fault, $\Delta t = 5\,\mathrm{s}$.: Drone 2 severs at $t = 17\,\mathrm{s}$, $5\,\mathrm{s}$ after the first. Peak error $0.446\,\mathrm{m}$, IAE $0.666\,\mathrm{m\,s}$, max sag $79.1\,\mathrm{mm}$, settle time $0.158\,\mathrm{s}$. The post-second-fault transient decays within the same envelope established from the first fault, consistent with $\rho < 1$.

c) V5 — dual fault, $\Delta t = 10\,\mathrm{s}$.: Extending the gap to $10\,\mathrm{s}$ lets the system fully re-settle. The second fault's IAE is $0.741\,\mathrm{m\,s}$ ($+11\%$ over V4) reflecting higher cruise speed at $t = 22\,\mathrm{s}$, not degraded recovery. Max sag $87.4\,\mathrm{mm}$; sub-tick settle.

d) V6 — full stack.: V6 repeats V4 with MPC $+L_1$+reshape active. First-fault sag is $69.9\,\mathrm{mm}$ (-12% vs V4), confirming the stack preserves the envelope while delivering incremental benefit.

Figure 7 shows the fault-centered timeline for V4 with FF-on (solid) vs FF-off (dashed) on the same wind seed: (a) tracking-error norm, (b) altitude error exposing the sag differential, (c) Lyapunov proxy $V(t) = \xi^\top P_v \xi$, allowing direct visual inspection of the bounded jump (Theorem 5) and the inter-event contraction (Theorem 6) alongside the causal mechanism. V3, V5, FF-held-constant, and per-drone thrust traces are in the supplementary package. Empirical fits on V4 and V5 give peak jump $\hat{\chi}_{\max} = 1.1 \times 10^{-4}$ (V4), 1.3×10^{-5} (V5)—in units of the dimensionless proxy V, not physical units—and dwell-cycle contraction $\hat{\rho}_{\mathrm{V4}} = 0.20$, $\hat{\rho}_{\mathrm{V5}} = 0.045$, both well below the $\rho < 1$ sufficiency bound (Table III).

The deterministic effect-size estimates are in Table IX and Figure 7; statistical amplification across paired CRN seeds is follow-up work.

E. Tension Feedforward Ablation

The P2-A campaign provides the strongest quantitative evidence for (31) as the causal recovery mechanism. We disable $T_i^{\mathrm{ff}} = T_i$ in the QP layer with all other parameters held fixed and repeat V3, V4, V5.

Disabling the feed-forward raises cruise RMSE by 34–39% and inflates peak sag by 3.6–4.0×. The degradation appears even pre-fault: without the feed-forward, the outer PD must reject the quasi-static sag from payload weight and centripetal acceleration, raising the steady-state floor that γ_η must absorb. Post-fault, survivors cannot absorb the redistribution at the plant's native rate; the transient decays on the pendulum damping timescale rather than the control-tick scale. Table IX summarises and Figure 7 shows the V4 time-series comparison.

TABLE IX
TENSION FEED-FORWARD ABLATION (P2-A). CRUISE-WINDOW RMSE (M) AND PEAK POST-FAULT ALTITUDE SAG (MM) WITH FF ENABLED AND DISABLED, PER FAULT SCHEDULE.

	RMSE (m)			Peak sag (mm)		
Variant	FF on	FF off	Δ%	FF on	FF off	×
V3	0.324	0.432	+34 %	86.8	309	3.6×
V4	0.328	0.457	+39 %	95.2	358	3.8×
V5	0.324	0.444	+37 %	90.4	359	4.0×

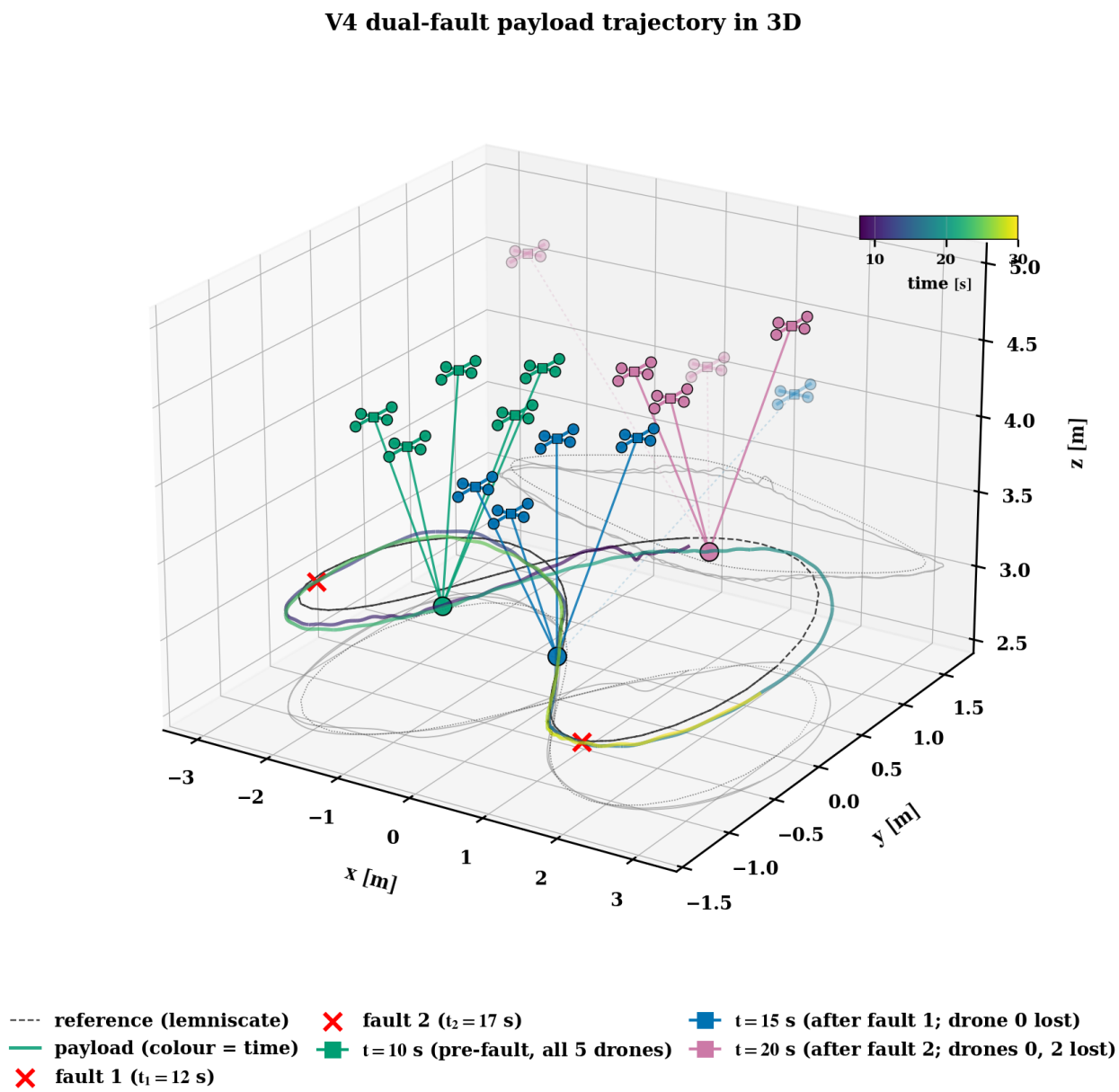


Fig. 6. V4 dual-fault payload trajectory. Time-coloured curve: payload over $t \in [8, 30]$ s. Dashed: lemniscate reference. Red crosses: fault events at t_1=12 s (drone 0) and t_2=17 s (drone 2). Formation snapshots at $t \in \{10, 15, 20\}$ s show pre-fault, post-fault 1, and post-fault 2 geometries (solid: taut, dashed: severed). Grey shadows: top-down and side-view projections.

F. Dwell-Time Boundary Characterisation

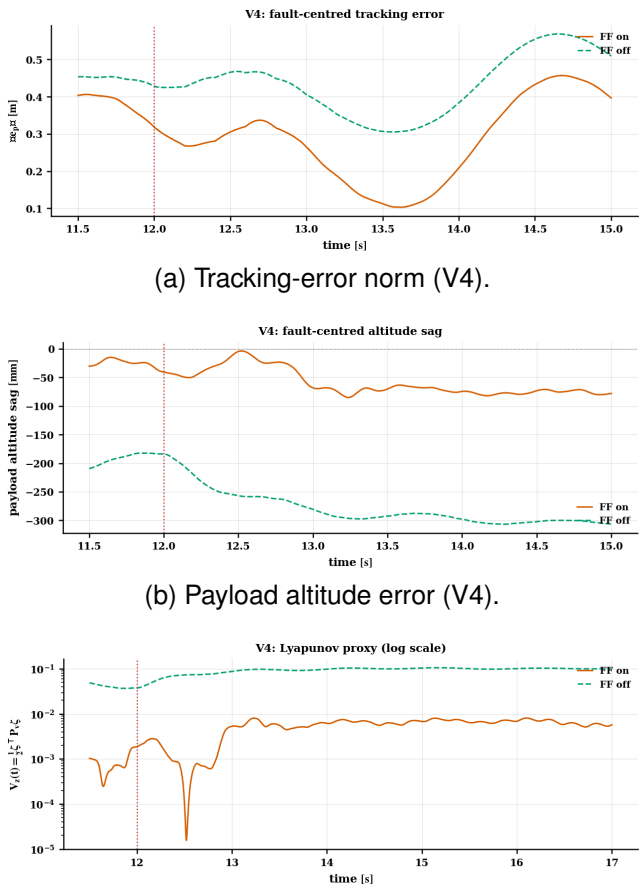
(11) asserts $\rho = e^{-\alpha_{\min}\tau_{\text{pend}}} < 1$ for any $\tau_d > \tau_{\text{pend}}$. V4 and V5 confirm two operating points inside the guaranteed regime but do not characterize behavior as τ_d approaches or falls below τ_{pend}. We close this gap with a boundary sweep on V4 across $\tau_d/\tau_{\text{pend}} \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$ at seed 42.

For each point we compute the per-fault peak altitude excursion $\hat{e}^{\max}_{p,z}(t_k)$ relative to a 0.2-second DC-offset window ending at t_k (removing the post-fault equilibrium shift), then form $\hat{\rho}(\tau_d) = \hat{e}^{\max}_{p,z}(t_2)/\hat{e}^{\max}_{p,z}(t_1)$. Figure 9 reports the result.

The empirical result is stronger than the hypothesis demands: $\hat{\rho} < 1$ at every tested dwell, including the sub-threshold $\tau_d = 0.5\tau_{\text{pend}}$ where $\hat{\rho} \approx 0.63$. The ratio falls to $\hat{\rho} \approx 0.14$ at $\tau_d = 1.5\tau_{\text{pend}}$. The non-monotonicity at $\tau_d = 2.0\tau_{\text{pend}}$ reflects the trajectory phase the second fault hits (centripetal peak vs trough); a fault-phase sweep would resolve it. The fault-2 peak is consistently below fault-1 across the sweep: the identity converts measured tension into a thrust increment at the plant's native rate regardless of dwell. H2 remains a structurally motivated bound; the empirical map shows contraction margin persists below it on this trajectory family.

G. Parameter Mismatch and Adaptive Augmentation

P2-B evaluates robustness to payload-mass mismatch and the L_1 benefit (Section V-A, C3-L_1). We sweep $m_L \in \{2.5, 3.0, 3.5, 3.9\}$ kg (the PD design range, distinct from the



(a) Tracking-error norm (V4).

(b) Payload altitude error (V4).

(c) Lyapunov proxy $V = \xi^\top P_v \xi$ (V4).

Fig. 7. Fault-centered causal ablation on V4, FF-on (solid) vs FF-off (dashed, P2-A) at seed 42. *(a)* tracking-error norm; *(b)* altitude error; *(c)* Lyapunov proxy $V(t) = \xi^\top P_v \xi$, exhibiting the bounded jump of Lemma 5 and the inter-event contraction of Lemma 6.

10 kg V1–V6 payload) with $q^\star$ fixed at 3.0 kg and compare FF-on, FF-off, FF-off+L_1. All runs use the fault-free trajectory to isolate mismatch effects. The L_1 evidence is obtained in rescue mode (FF off, L_1 alone), which isolates the L_1 contribution; main-mode evaluation at the canonical 10 kg point is follow-up work.

FF-on is robust across the range: RMSE varies $< 1.9\%$, confirming the identity pre-compensates the per-drone load change without explicit mass estimation (Theorem 9(i)). FF-off degrades approximately linearly, reaching 0.350 m RMSE at $m_L = 3.9$ kg vs 0.341 m at the nominal. FF-off+L_1 recovers $\sim 40\%$ of the mismatch-induced sag.

The single-axis sweep of Figure 10 is one projection of the four-axis claim of Theorem 9. The full response surface over $(m_L, k_s, c_{\text{drag}}, \bar{w})$ on $n = 100$ CRN-paired seeds is identified as a campaign extension.

a) L_1 adaptive-gain stability map.: Proposition 1 gives the discrete-time sufficient bound $\Gamma < 2/(T_s p_{22}) \approx 4.76 \times 10^5$. We sweep $\Gamma \in \{0.5, 2, 10, 30, 50, 80\} \times 10^3$ ($\Gamma/\Gamma^\star \in [1.1 \times 10^{-3}, 0.17]$) on a fault-free 20 s mass-mismatch run ($m_L = 3.9$ kg vs $q^\star = 3.0$ kg, FF off). Each drone runs its own L_1 update with the swept Γ, with no shared state or orchestrator. Altitude RMSE (Figure 11) drops from 101 mm at $\Gamma = 500$ to a minimum of 88.5 mm at $\Gamma \approx 10^4$,

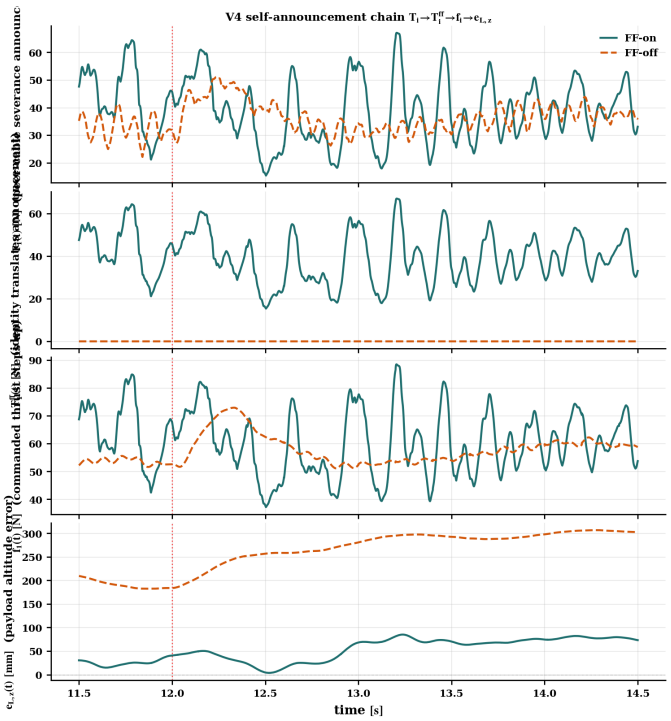


Fig. 8. Self-announcement causal chain (V4, fault 1, drone 0 severed). Rows top to bottom: surviving rope tension $T_1(t)$, feed-forward command $T_1^{\text{ff}}(t)$, commanded thrust $f_1(t)$, altitude error $e_{L,z}(t)$. Teal solid: baseline (FF-on); orange dashed: P2-A ablation (FF-off). With the identity active, T_1^{ff} tracks T_1 at the next tick, thrust steps proportionally, and altitude error stays bounded; with it disabled the thrust step is absent and altitude error grows by ~ 300 mm.

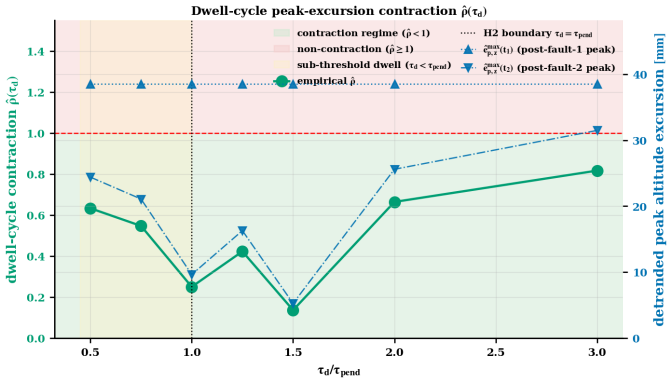


Fig. 9. Dwell-cycle peak-excursion contraction $\hat{\rho}(\tau_d)$ versus normalised dwell $\tau_d/\tau_{\text{pend}}$, V4 schedule, seed 42. Horizontal dashed: $\hat{\rho} = 1$; vertical dotted: H2 boundary. $\hat{\rho}$ remains below unity across the full sweep, including sub-threshold dwell.

then rises to 91.4 mm at $\Gamma = 8 \times 10^4$. Peak sag follows the same shape, 130 mm down to $\sim$ 113 mm. The late-vs-early variance ratio stays in $[0.36, 0.37]$; no instability is observed. The sweep stays inside the contraction-guarantee region ($\max_\Gamma \Gamma/\Gamma^\star \approx 0.17$); the smooth shape supports $\Gamma^\star$ as a non-vacuous sufficient bound. Probing past $\Gamma^\star$ is follow-up work.

H. Extension Characterization and Binding-Regime Evidence

P2-C and P2-D probe the MPC tension-ceiling (C3-MPC) and reshape supervisor (C3-Reshape) on V4. Both yield null

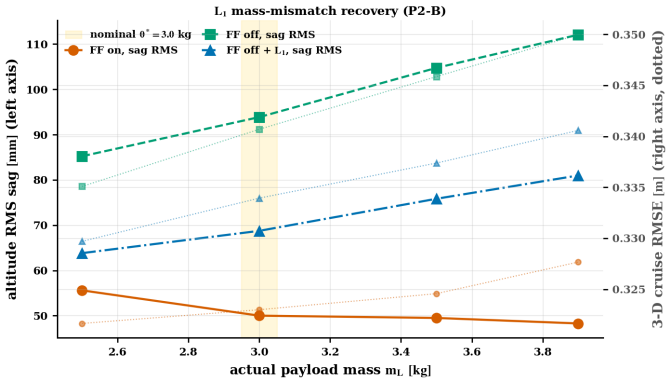


Fig. 10. P2-B payload-mass-mismatch sweep. Cruise-window 3-D RMSE versus $m_L \in \{2.5, 3.0, 3.5, 3.9\}$ kg with $q^\star = 3.0$ kg for FF-on, FF-off, and FF-off+L_1. FF-off+L_1 recovers $\sim 40\,\%$ of the FF-off-minus-FF-on sag gap.

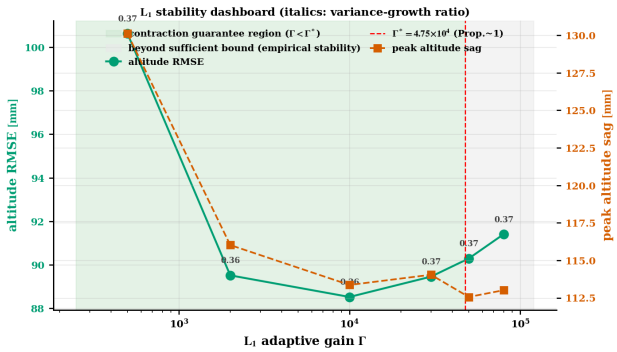


Fig. 11. L_1 adaptive-gain stability dashboard. Left axis (circles): altitude RMSE vs Γ. Right axis (squares, dashed): peak altitude sag. Italics: late-vs-early error-standard-deviation ratio. Green band: contraction-guarantee region $\Gamma < \Gamma^\star \approx 4.76 \times 10^5$ (Proposition 1). FF-off rescue mode, $m_L = 3.9$ kg, no faults.

findings on the canonical family, complemented by binding-regime probes for the operating conditions under which each extension activates.

a) P2-C — MPC tension ceiling.: We sweep $T_{\max} \in \{60, 70, 80, 90, 100\}$ N against the unconstrained baseline (Table X).

Remark 5 (NF2: MPC ceiling non-binding in canonical regime)**.** *Peak tension during the fault transient is dominated by the pre-fault dynamic spike from the lemniscate's centripetal acceleration ($\approx$ 42 N on survivors immediately before the first fault, resolving within one pendulum period); this spike is independent of the ceiling. The ceiling targets the post-fault quasi-static regime: at the 5→3 transition with* 10 kg *payload,* $m_L g/3 \approx 32.7$ N*, well below all tested ceilings. Lowering the ceiling from* 100 N *to* 60 N *produces no measurable change (NF2). The ceiling becomes binding when the post-fault load approaches* $T_{\max}$*—heavier payload, lower ceiling, or fewer survivors. Proposition 2 and Section V-D preserve Theorem 7 whether or not the ceiling is binding.*

b) P2-D — Formation reshape.: We sweep $T_{\text{ref}} \in \{8, 10, 12\}$ s with and without reshape on V4, plus the aggressive-period binding-regime probe at $T_{\text{ref}} = 6$ s (E8). Peak post-fault rope tension is identical with reshape on and off at every tested period: 330.2 N, 191.4 N, 153.8 N, 87.9 N at $T_{\text{ref}} = 6, 8, 10, 12$ s respectively.

Remark 6 (NF3: Reshape benefit zero across periods and aggressive-period probe)**.** *The reshape supervisor targets the quasi-static-hover tension distribution under the symmetric-load-sharing model of Theorem 11. At the tested periods, centripetal acceleration* $a_c = (2\pi/T_{\text{ref}})^2 A \approx 0.82$–$3.29$ m s^{-2} *continuously modulates the per-rope distribution, and this dynamic modulation dominates the hover-equilibrium asymmetry the supervisor targets—so high centripetal load, far from making the asymmetry binding, makes it* non-*binding. The quasi-static assumption of Theorem 11 therefore fails to bind at any tested period, including the aggressive* $T_{\text{ref}} = 6$ s *probe (E8) for which the linearised theorem a priori predicted a* $\sim 25.8\%$ *reduction; the measured benefit there is* 0.0%*. Pushing reshape into a binding regime would require a lower-bandwidth reference family or an extended objective accounting for the dynamic load. Theorem 11 and Section V-D preserve the envelope outside the binding regime.*

TABLE X
MPC TENSION-CEILING SWEEP (P2-C, NF2). PEAK ROPE TENSION AND FRACTION OF MISSION TIME OVER THE COMMANDED CEILING $T_{\max}$, V4 FAULT SCHEDULE.

Ceiling $T_{\max}$	Peak T	Peak post-fault	Time over ceiling
60N	104.6N	87.9N	17.52%
70N	104.6N	87.9N	6.50%
80N	104.6N	87.9N	1.60%
90N	104.6N	87.9N	0.34%
100N	104.6N	87.9N	0.12%
Baseline	104.6N	87.9N	—

I. Trajectory-Phase Sensitivity of Recovery

V4 fixes $t_1 = 12$ s. We ask whether the recovery envelope depends on *where in the lemniscate cycle* the first fault hits, i.e. whether t_1=12 s is fortuitous or representative. We sweep $t_1 \in \{8, 10, 12, 14, 16\}$ s on V4 with $\Delta t = 5$ s and the same fault drones, so the first fault lands at five distinct phases.

Figure 12 shows peak post-fault sag and 3-D cruise RMSE. Peak sag varies from 12.0 mm ($t_1 = 14$ s, low-acceleration phase) to 55.4 mm ($t_1 = 12$ s, near a centripetal peak), a 4.6× ratio exceeding the 3.6–4.0× FF-ablation magnitude. 3-D cruise RMSE varies modestly, 313 mm–336 mm ($\sim 7\%$ range): cruise-window tracking is robust to fault phase even when per-event peaks are not. Canonical $t_1 = 12$ s sits at the upper end of the sag distribution—a worst-case phase, not a fortuitous one—which strengthens rather than weakens the recovery claims.

J. H2 Sub-Threshold Dwell Probe

Figure 13 shows the H2 violation probe: the sub-threshold dwell $\tau_d = 0.5\,\tau_{\text{pend}} = 1.12$ s on V4, with the Lyapunov proxy overlaid against the canonical V4 run. The sub-threshold trajectory is bounded and $\hat{\rho}$ (Section VI-F and Figure 9) is below unity even here, showing the identity delivers contraction margin beyond H2's sufficiency. The probe illustrates the

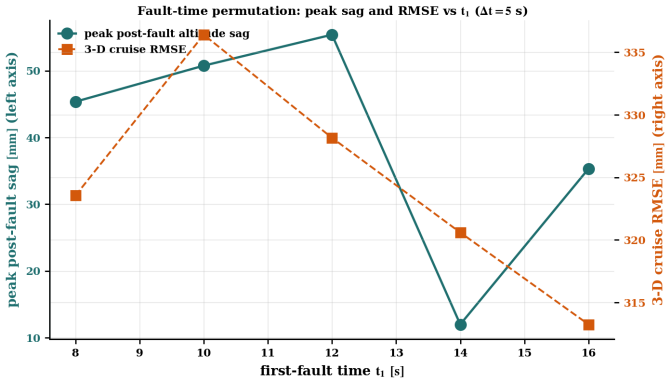


Fig. 12. Fault-time permutation. V4 dual-fault, $\Delta t = 5\,\mathrm{s}$, drones 0, 2; $t_1 \in \{8, 10, 12, 14, 16\}$ s. Left axis: peak post-fault altitude sag. Right axis (dashed): 3-D cruise RMSE. The canonical $t_1 = 12$ s is a worst-case within the sweep.

absence of an empirical H2 failure on this family; constructing a regime that empirically violates contraction is follow-up work.

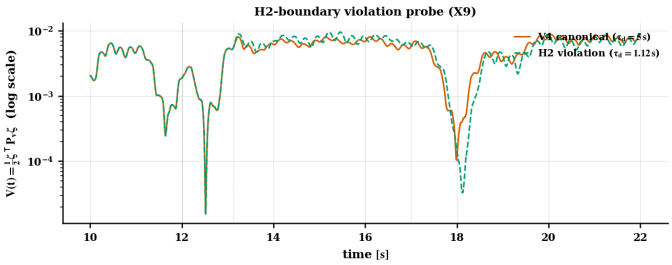


Fig. 13. H2 sub-threshold dwell probe. Lyapunov proxy $V(t) = \xi^\top P_v \xi$ at $\tau_d = 0.5\,\tau_{\mathrm{pend}} = 1.12\,\mathrm{s}$ overlaid with the canonical V4 trajectory ($\tau_d = 5\,\tau_{\mathrm{pend}}$). Red ticks: fault events. The sub-threshold trajectory remains bounded; $\hat{\rho}$ on the detrended peak-excursion definition of Section VI-F sits below unity (Figure 9).

K. Actuator-Margin and Recovery-Time Audit

A reviewer concern with cascade architectures is that the controller may rely on actuator saturation to absorb post-fault loads. We audit this with the per-drone maximum commanded thrust ratio $f_i(t)/f_{\max}$ over $[8, 30]\,\mathrm{s}$ for V3–V6. The variant maxima are 0.73 on V3, V4, V5 and 0.79 on V6, all attained on drone 4, with zero cruise-window time above $0.9\,f_{\max}$ on every variant. The V3–V5 maxima coincide because the three variants share the first-fault schedule (drone 0 severs at $t = 12\,\mathrm{s}$) and the simulations are deterministic on a common wind seed: the variant peak is drone 4's transient at $t = 12.68\,\mathrm{s}$, $0.68\,\mathrm{s}$ into the fault-1 recovery, and the second-fault spike (V4: $t = 17\,\mathrm{s}$, V5: $t = 22\,\mathrm{s}$) lands on a different drone with the trajectory at a different phase and stays below the fault-1 peak on drone 4. V6 reaches 0.79 because the full-stack extensions (L_1 + MPC + reshape) shift the altitude command. Recovery is not actuator-limited on any variant.

a) Spike-then-relax thrust dynamics on V4.: The variant-level maximum hides the intra-fault dynamic that directly evidences H2 on the actuator channel. Table XI reports, per V4 survivor ($i \in \{1, 3, 4\}$), the within-1 s peak commanded thrust at each fault and the settled mean 2–4 s after. Survivors show a 1.18–1.79× peak-over-settled overshoot, decaying to a new steady state within $\sim \tau_{\mathrm{pend}} \approx 2.24\,\mathrm{s}$. Steady-state thrust per survivor rises +25 to +34% vs the pre-fault baseline; the survivor sum settles near $179\,\mathrm{N}$ at $t \in [27, 30]\,\mathrm{s}$, balancing $m_L g + 3m_{\mathrm{drone}} g \approx 142\,\mathrm{N}$ plus centripetal/tilt overhead—an independent check of the post-fault load redistribution. This is the actuator-channel signature of the same recovery the Lyapunov-proxy contraction of Section VI-D reports for the state error.

TABLE XI
V4 SPIKE-THEN-RELAX THRUST DYNAMICS ON THE THREE SURVIVING DRONES. PEAK: MAX WITHIN 1 S OF EACH FAULT; SETTLED: MEAN 2–4 S AFTER THE FAULT. Δ: STEADY-STATE INCREASE ($t \in [27, 30]$ S) OVER THE PRE-FAULT $[10, 12]$ S BASELINE.

Drone	Fault	Pre (N)	Peak (N)	Settled (N)	Peak/Sett. ratio
1	f1 (t_1=12 s)	59.3	82.2	62.1	1.32×
1	f2 (t_2=17 s)	62.1	97.2	69.1	1.41×
3	f1	40.7	53.4	40.1	1.33×
3	f2	40.1	77.8	52.2	1.49×
4	f1	41.7	110.0	61.4	1.79×
4	f2	61.4	63.3	53.7	1.18×
Steady-state ($t \in [27, 30]$ s) vs pre-fault baseline:					
1	—	53.3	—	75.1	$\Delta = +34\,\%$
3	—	42.8	—	53.0	$\Delta = +25\,\%$
4	—	42.4	—	50.8	$\Delta = +26\,\%$

Table XII reports per-fault metrics beyond cruise RMSE: peak error norm, peak sag, recovery time t_{rec} to within $\varepsilon = 0.35\,\mathrm{m}$ for $0.3\,\mathrm{s}$ continuous, and IAE over τ_{pend} post-fault. t_{rec} is bounded by the next fault's arrival in dual-fault variants; IAE captures the integrated per-event cost.

TABLE XII
PER-FAULT RECOVERY METRICS. t_{rec} THRESHOLD $0.35\,\mathrm{m}$ (3-D, $0.3\,\mathrm{s}$ CONTINUOUS HOLD); IAE INTEGRATED OVER τ_{pend} OR THE GAP TO THE NEXT FAULT, WHICHEVER IS SHORTER.

Variant	Fault	$t^\star$	peak $\|e\|$	peak sag	t_{rec}	IAE
V3	1	12.0s	481mm	85mm	0.00s	0.52m·s
V4	1	12.0s	481mm	85mm	0.00s	0.52m·s
V4	2	17.0s	457mm	89mm	0.32s	0.70m·s
V5	1	12.0s	481mm	85mm	0.00s	0.52m·s
V5	2	22.0s	469mm	90mm	1.58s	0.81m·s
V6	1	12.0s	481mm	73mm	0.00s	0.51m·s
V6	2	17.0s	460mm	72mm	0.33s	0.70m·s

L. Reproducibility and Claim-to-Evidence Map

Remark 7 (Simulation protocol and seed manifest)**.** *V1–V6 use a fixed Dryden seed 42, producing bit-exact output across runs at identical Drake and library versions. Inter-variant RMSE differences are therefore attributable to controller configuration and fault schedule alone. Post-hoc inspection confirms the principal (+x) wind component achieves a 2σ exceedance within $[8, 12]$ s, making the realization conservative relative to the mean. A Monte Carlo sweep would underestimate the worst case by averaging over favorable realizations.*

Ablations (P2-A through robustness sweep) are deterministic point estimates at seed 42; CRN-paired replication yielding bootstrap CIs is follow-up work. The seed manifest (`seed_manifest.csv`: `seed_wind`, `seed_init`, `seed_sensor`, `seed_fault`, `seed_solver`) is archived alongside simulation outputs.

The audit trail: C1 is supported by Tables VI and VII; C2 by Figures 7 and 9; C3 by Figure 11; FF causality by Table IX and Figure 8; actuator margin and recovery time by Tables XI and XII (with the variant-level maxima reported inline in Section VI-K); operating-boundary characterisation by Section VI-J.

VII. Discussion

The campaign of Section VI establishes a single architectural commitment: when the structural transition self-announces through actuator-side measurements on the plant's native timescale, the announcement-detection-reconfiguration timeline of classical active fault-tolerant pipelines is unnecessary, and a single decentralized identity $T_i^{\mathrm{ff}} = T_i$ on the per-drone altitude channel suffices. The five-drone, ten-kilogram dual-fault scenario is one point on the feasibility surface (Figure 2). We discuss what the campaign establishes and where its boundary leaves open questions.

A. Audit of the Three Contributions

Theorem 1 (C1) is defended in two layers. The domain-gate audit (Table VI) certifies all six missions sit inside $\Omega_\tau^{\mathrm{dwell}}$ with non-trivial margin. The fidelity layer (Section VI-B) quantifies the per-rope dispersion the lumped scalar abstracts: cruise-window RMS residual $12.4\,\mathrm{N}$ on dual-fault variants, peaks $60\,\mathrm{N}$ at the most centripetally-loaded instants. The dispersion is informative rather than alarming: the $\mathcal{O}(\delta+\eta_{\max})$ guarantee bounds closed-loop trajectory deviation, not per-rope difference, and the identity reads each drone's *own* measured tension. A trajectory-level comparison against a parallel reduced-rope simulator is the principal extension of the C1 audit.

Theorem 7 (C2) is supported by three converging measurements: all six variants pass the pre-registered thresholds with dual-fault peak sag $95.2\,\mathrm{mm}$ vs $100\,\mathrm{mm}$ ceiling (Table VIII); fits on V4/V5 Lyapunov proxy give $\hat{\rho}_{\mathrm{V4}} = 0.20$, $\hat{\rho}_{\mathrm{V5}} = 0.045$; the dwell sweep shows $\hat{\rho} < 1$ across $\tau_d/\tau_{\mathrm{pend}} \in [0.5, 3.0]$ including the sub-threshold point. Proposition 1 (C3) is validated by the stability dashboard: across Γ spanning 2.4 decades, RMSE is unimodal with smooth degradation past $\Gamma^\star$—the signature of a sufficient-condition Lyapunov bound. The three contributions converge on a single causal anchor: the P2-A ablation (Table IX and Figure 8) raises cruise RMSE by 34–39% and inflates peak sag 3.6–$4.0\times$, isolating the identity as the mechanism on which C1, C2, and C3 jointly depend.

B. What the Architecture Changes

Existing decentralized cooperative-transport architectures address fault tolerance through one of three families. Active fault-tolerant pipelines [22] estimate, classify, and reassign on a detection-confirmation-reassignment timeline comparable to or longer than $\tau_{\mathrm{pend}} \approx 2.24\,\mathrm{s}$. Passive force-control and leader–follower architectures [23], [24] avoid inter-UAV exchange but presuppose fixed cable cardinality, with no guarantee across the constraint-set transition. Centralized and synchronous-distributed force allocation [1], [7]–[10] is structurally inapplicable: the per-tick global-state synchronization it requires is precisely what the constraint-cardinality transition breaks at the actuator-tick rate.

The commitment here is to read the transition off the actuator-side measurement and convert it directly into a thrust increment through $T_i^{\mathrm{ff}} = T_i$, eliminating the announcement-detection-reconfiguration timeline rather than substituting a faster version. The empirical signature is the per-fault recovery times in Table XII: $0.00\,\mathrm{s}$ on first faults (sub-tick) and $0.32\,\mathrm{s}$, $0.33\,\mathrm{s}$ on V4/V6 second faults, well inside one pendulum period. The four architectural properties of the introduction follow as consequences: *locality* by $\mathcal{I}_i$ (13); *passivity* by the absence of detector, classifier, or scheduler in the loop; *robustness to multi-agent failure* by induction on the fault counter, with the demonstrated $(N{=}5, m_L{=}10\,\mathrm{kg}, F{=}2)$ consuming 27% of the envelope (Figure 2); and *team-size independence*, since the per-drone law and lemma stack carry no N structurally and the dependence concentrates in $m_L g/N$ and $m_L g/(N{-}F)$.

The two null findings (NF2, NF3) are architectural evidence rather than failed extension claims. The MPC layer shows essentially identical peak tension $\approx 104.6\,\mathrm{N}$ across all ceilings (Table X) because the pre-fault dynamic spike dominates the quasi-static load the ceiling targets. The reshape supervisor returns 0.0% benefit at every tested period (Remark 6) because the dynamic centripetal load dominates the hover asymmetry Theorem 11 optimises against. Both nulls show the conventional active-FT machinery to be *structurally inactive* on the demonstrated family; the bookkeeping of Section V-D preserves Theorem 7 unchanged.

C. Boundary of the Architectural Category

Three regimes lie outside the analysis domain. Rapid multi-fault ($\tau_d < \tau_{\mathrm{pend}}$) voids Theorem 6; constructing a regime that empirically violates contraction is the natural completion of the dwell-boundary characterisation. Long slack excursions exit $\Omega_\tau^{\mathrm{dwell}}$, and QP fallback breaks D1; both are open to the same boundary search and benefit from a multi-fault extension of the dwell argument beyond pairwise composition.

A second class targets the linearisation residuals. Tube-MPC tightening folding the recovery-envelope overshoot into the QP RHS would convert the P2-C null into a positive result on spike-dominated regimes; an aggressive-period reshape with an objective extended to the dynamic centripetal load would similarly close the P2-D null; main-mode L_1 evaluation at the canonical $10\,\mathrm{kg}$ point would complete C3-L_1; and a four-axis $(m_L, k_s, c_{\mathrm{drag}}, \bar{w})$ response surface on CRN-paired seeds is the visualization counterpart of the same claim. A third class lies outside the architectural category itself—drone-side faults, soft cable failures, tension-sensor failures, payload-attitude dynamics—enumerated in Section V-D. These are not gaps in the present campaign.

Within this category, the campaign establishes that the identity $T_i^{\mathrm{ff}} = T_i$ delivers hybrid practical input-to-state stability with measured contraction margin across single and dual unannounced severances, that the identity is necessary in the precise sense P2-A isolates, and that the conventional active-FT machinery is structurally inactive on the trajectory family for which the identity is binding.

VIII. Conclusion

In this paper, we addressed cooperative aerial manipulation under unannounced cable severance through a single architectural commitment: the post-fault tension redistribution reaches each surviving drone through its own actuator-side measurement on the plant's native timescale. The per-drone altitude feed-forward identity $T_i^{\mathrm{ff}} = T_i$ converts that signal directly into thrust at the next control tick. We presented a three-layer decentralized cascade — outer-loop PD with anti-swing slot shift, single-step convex QP on the tilt-and-thrust envelope, and geometric attitude inner loop — in which this identity is the sole non-standard element. This commitment is supported by three constructive certificates: a taut-cable reduction theorem that bounds the closed-loop deviation between the distributed-rope and lumped-rope plants by $O(\delta + \eta_{\max})$ on a slack-excursion-bounded admissibility set; a hybrid practical-ISS recovery envelope that yields a closed-form per-cycle Lyapunov contraction $\rho \in (0, 1)$ across each unannounced severance; and a closed-form discrete-time stability bound that certifies the optional L_1 adaptive altitude augmentation, removing the dominant hyper-parameter of the architecture from empirical tuning.

We validated the proposed controller in high-fidelity Drake multibody simulation through a six-variant capability sequence and four targeted ablation campaigns at $N = 5$, $m_L = 10\,\mathrm{kg}$, and $4\,\mathrm{m\,s^{-1}}$ Dryden wind. The feed-forward ablation pins the identity as the causal recovery mechanism: removing it inflates cruise RMSE by 34–39 % and post-fault sag by 3.6–4.0×, while the intact controller absorbs each severance through a thrust spike that settles within one payload-pendulum period and never reaches the saturation hazard line. The measured per-cycle Lyapunov contraction sits well below the $\rho < 1$ sufficiency bound and remains below unity even at sub-threshold dwell. The per-drone information set carries no peer state and no fault flag; the recovery envelope therefore extends by induction to $F \leq N-2$ as long as actuator margin remains, and the demonstrated configuration consumes only a fraction of that margin. The architecture eliminates the announcement-detection-reconfiguration timeline entirely: in this regime, the identity *is* the response.

For future work, we identify four directions within the architectural category. First, a parallel reduced-rope simulator that audits the reduction theorem at the trajectory level rather than at the domain-gate level. Second, a multi-fault generalization of the dwell-time argument beyond pairwise composition, informed by the empirical sub-threshold contraction we observed. Third, a tube-MPC tightening that folds the post-fault overshoot of the recovery envelope into the constraint right-hand side, converting the spike-dominated null finding into a positive result on trajectories where the dynamic spike is the binding constraint. Fourth, a head-to-head comparison against active fault-tolerant baselines at the actuator-tick rate, isolating the detection-loop contribution to recovery latency. Drone-side faults, soft cable failures, tension-sensor failures, and payload-attitude dynamics lie outside the architectural category; each requires its own announcement channel and is properly the subject of a different paper. The cable-severance instance is the certification vehicle, not the contribution. The contribution is the category itself: for self-announcing structural transitions on the plant's native timescale, the announcement is the response, and detection is unnecessary.

References

[1] N. Michael, J. Fink, and V. Kumar, "Cooperative manipulation and transportation with aerial robots," *Autonomous Robots*, vol. 30, no. 1, pp. 73–86, 2011.

[2] J. Fink, N. Michael, S. Kim, and V. Kumar, "Planning and control for cooperative manipulation and transportation with aerial robots," *The International Journal of Robotics Research*, vol. 30, no. 3, pp. 324–334, 2011.

[3] M. Bernard, K. Kondak, I. Maza, and A. Ollero, "Autonomous transportation and deployment with aerial robots for search and rescue missions," *Journal of Field Robotics*, vol. 28, no. 6, pp. 914–931, 2011.

[4] A. Tagliabue, M. Kamel, R. Siegwart, and J. Nieto, "Robust collaborative object transportation using multiple MAVs," *The International Journal of Robotics Research*, vol. 38, no. 9, pp. 1020–1044, 2019.

[5] B. Brogliato, *Nonsmooth Mechanics: Models, Dynamics and Control*, 3rd ed., ser. Communications and Control Engineering. Cham, Switzerland: Springer International Publishing, 2016.

[6] R. Goebel, R. G. Sanfelice, and A. R. Teel, *Hybrid Dynamical Systems: Modeling, Stability, and Robustness*. Princeton, NJ, USA: Princeton University Press, 2012.

[7] Z. Wang and C. J. Ong, "Distributed model predictive control of linear discrete-time systems with local and global constraints," *Automatica*, vol. 81, pp. 184–195, Jul. 2017.

[8] G. Li and G. Loianno, "Nonlinear model predictive control for cooperative transportation and manipulation of cable suspended payloads with multiple quadrotors," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, Oct. 2023.

[9] A. Pallar, G. Li, M. Sarvaiya, and G. Loianno, "Optimal trajectory planning for cooperative manipulation with multiple quadrotors using control barrier functions," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, May 2025.

[10] Y.-H. Su, P. Bhowmick, and A. Lanzon, "A robust adaptive formation control methodology for networked multi-UAV systems with applications to cooperative payload transportation," *Control Engineering Practice*, Sep. 2023.

[11] T. Lee, M. Leok, and N. H. McClamroch, "Geometric tracking control of a quadrotor UAV on SE(3)," in *49th IEEE Conference on Decision and Control (CDC)*. Atlanta, GA, USA: IEEE, 2010, pp. 5420–5425.

[12] K. Sreenath, N. Michael, and V. Kumar, "Trajectory generation and control of a quadrotor with a cable-suspended load — A differentially-flat hybrid system," in *2013 IEEE International Conference on Robotics and Automation (ICRA)*. Karlsruhe, Germany: IEEE, 2013, pp. 4888–4895.

[13] T. Lee, "Geometric control of quadrotor UAVs transporting a cable-suspended rigid body," *IEEE Transactions on Control Systems Technology*, vol. 26, no. 1, pp. 255–264, Jan. 2018.

[14] T. Lee, K. Sreenath, and V. Kumar, "Geometric control of cooperating multiple quadrotor UAVs with a suspended payload," in *52nd IEEE Conference on Decision and Control (CDC)*. Firenze, Italy: IEEE, 2013, pp. 5510–5515.

[15] M. Sharma and S. Sundaram, "A geometric control approach for multi-UAV cooperative payload transfer," *Nonlinear Dynamics*, Jun. 2023.

[16] C. Cao and N. Hovakimyan, "Design and analysis of a novel L1 adaptive control architecture with guaranteed transient performance," *IEEE Transactions on Automatic Control*, vol. 53, no. 2, pp. 586–591, Mar. 2008.

[17] N. Hovakimyan and C. Cao, *L1 Adaptive Control Theory: Guaranteed Robustness with Fast Adaptation*, ser. Advances in Design and Control. Philadelphia, PA, USA: SIAM, 2010.

[18] Z. Wu, S. Cheng, P. Zhao, A. Gahlawat, K. A. Ackerman, A. Lakshmanan, C. Yang, J. Yu, and N. Hovakimyan, "$\mathcal{L}_1$quad: $\mathcal{L}_1$ adaptive augmentation of geometric control for agile quadrotors with performance guarantees," *IEEE Transactions on Control Systems Technology*, Mar. 2025.

[19] M. Blanke, M. Kinnaert, J. Lunze, and M. Staroswiecki, *Diagnosis and Fault-Tolerant Control*, 3rd ed. Berlin, Heidelberg, Germany: Springer, 2015.

[20] Y. Zhang and J. Jiang, "Bibliographical review on reconfigurable fault-tolerant control systems," *Annual Reviews in Control*, vol. 32, no. 2, pp. 229–252, 2008.

[21] R. J. Patton, "Fault-tolerant control: The 1997 situation," in *IFAC Symposium on Fault Detection, Supervision and Safety for Technical Processes (SAFEPROCESS'97)*. Hull, United Kingdom: IFAC, Aug. 1997, pp. 1033–1055, plenary survey.

[22] X. Liang, Z. Su, W. Zhou, G. Meng, and L. Zhu, "Fault-tolerant control for the multi-quadrotors cooperative transportation under suspension failures," *Aerospace Science and Technology*, vol. 119, p. 107139, 2021.

[23] A. Tagliabue, M. Kamel, S. Verling, R. Siegwart, and J. Nieto, "Collaborative transportation using mavs via passive force control," in *2017 IEEE International Conference on Robotics and Automation (ICRA)*, May 2017, pp. 5766–5773.

[24] P.-X. Wu, H.-A. Hung, C.-C. Yang, and T.-H. Cheng, "Cooperative transportation of drones without inter-agent communication," in *2020 American Control Conference (ACC)*, July 2020, pp. 677–682.

[25] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, "OSQP: an operator splitting solver for quadratic programs," *Mathematical Programming Computation*, vol. 12, no. 4, pp. 637–672, 2020.

[26] Tedrake, Russ and the Drake Development Team, "Drake: Model-based design and verification for robotics," 2019, available at https://drake.mit.edu. [Online]. Available: https://drake.mit.edu